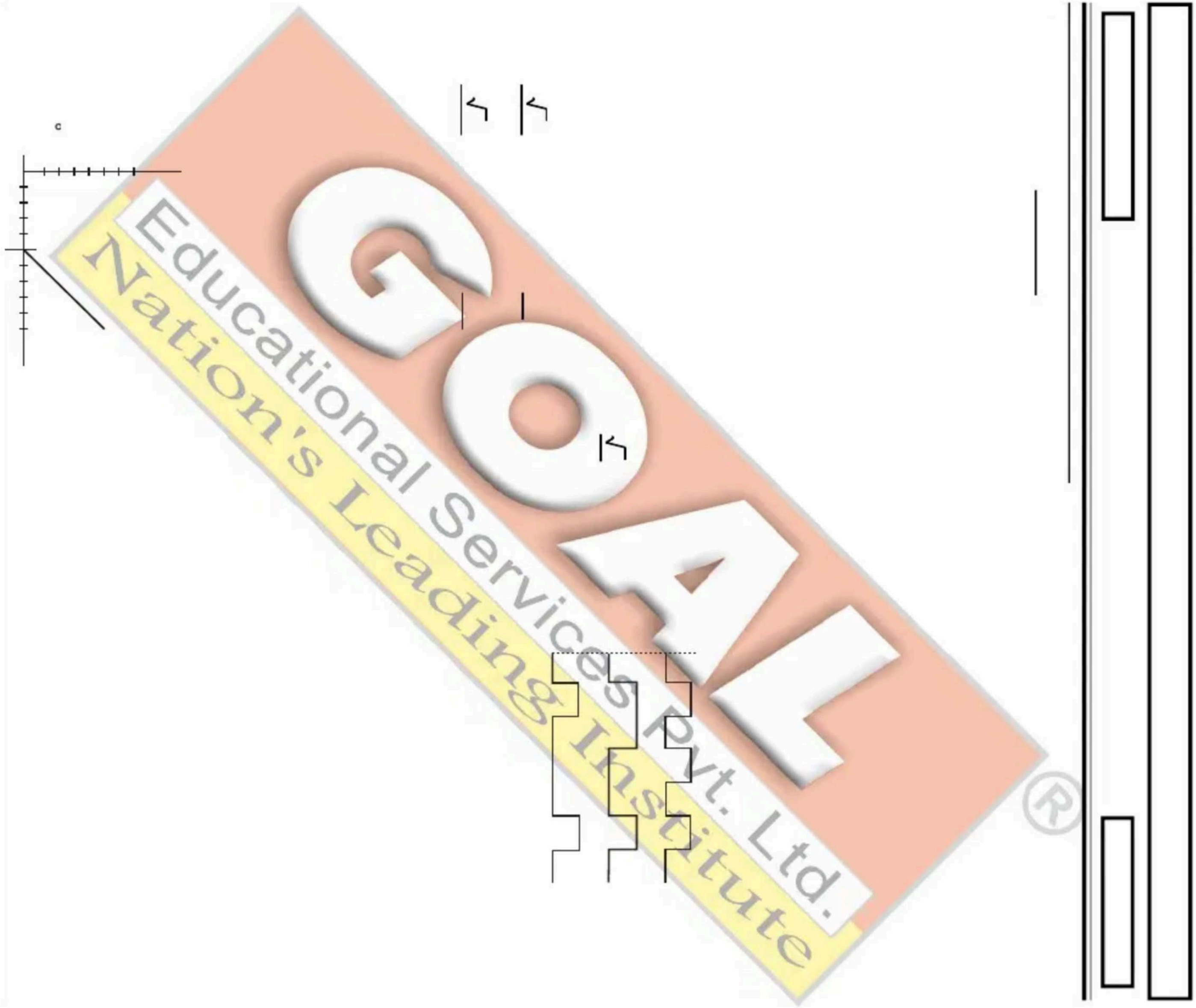
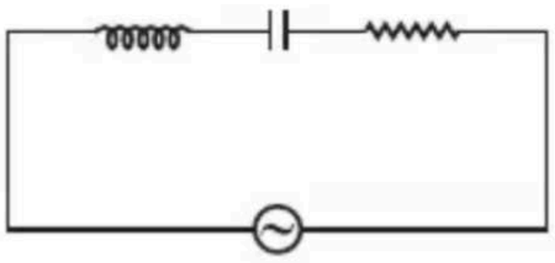
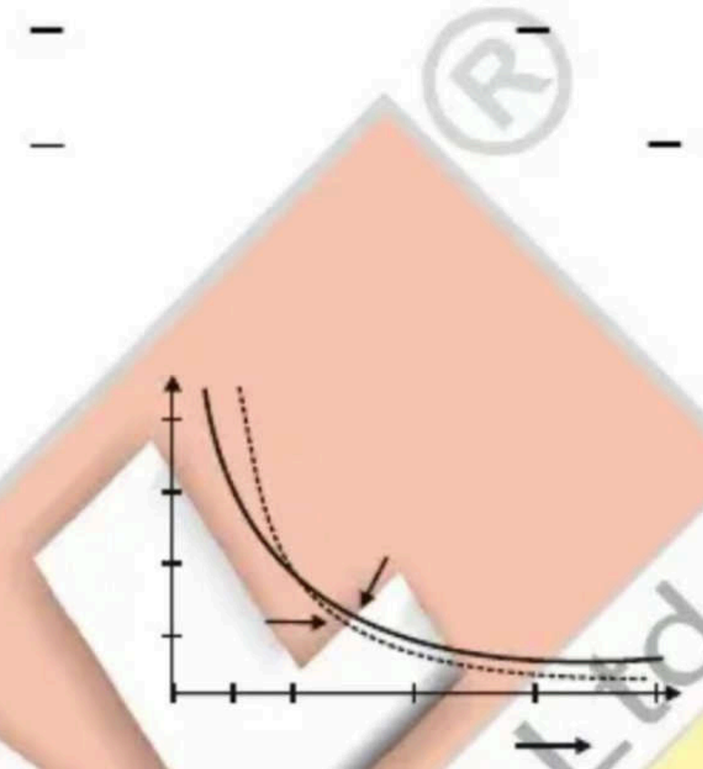
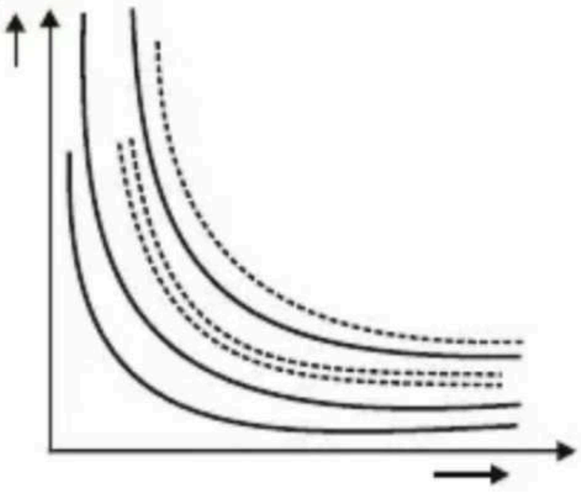
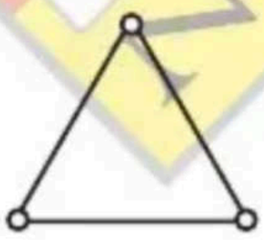

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28. The velocity of a particle is $v = v_0 + gt + ft^2$. If its position is x = 0 at t = 0, then its displacement after time (t = 2) sec. is :

(1) $2v_0 + 4g + \frac{3}{8}f$ (2) $2v_0 + 2g + \frac{8}{3}f$

(3) $v_0 + 4g + \frac{8}{3}f$ (4) $2v_0 + 2g + \frac{4}{3}f$

29. A hot wire of copper is stretched at a temperature of 200°C between two fixed walls. At what temperature will the wire break when it is cooled ? The breaking stress of copper is 4×10^8 N/m^2. Young's modulus of copper = 12×10^{10} N/m^2, coefficient of linear expansion of copper = 2×10^{-5}/°C.

(1) 33.33 °C (2) 67 °C

(3) 21.2 °C (4) 82 °C

30. Pressure at the bottom of tank of water is 4P, where P is atmospheric pressure. If the water is drawn out till the level of water is lowered by one-third, then the pressure at the bottom of the tank is :

(1) 2 P (2) $\frac{7P}{3}$

(3) 3P (4) $\frac{10P}{3}$

31. A ray of light passes through an equilateral prism such that the angle of incidence is equal to the angle of emergence and the latter is equal to $\frac{2}{3}$, the angle of prism. The angle of deviation is x°. Find the value of x.

(1) 30° (2) 25° (3) 15° (4) 20°

32. An energy of 18 eV is given to an electron in 3rd orbit of H-atom. Then, find its final energy when it comes out of H-atom.

(1) 14.67 eV (2) 16.49 eV

(3) 11.26 eV (4) 19.51 eV

33. In a Young's double slit experiment, d = 1 mm and D = 120 cm. It is found that 6th bright fringe is at a distance of 3.96 mm from 1st dark fringe of fringe pattern. The wavelength of light used is :

(1) 6000 Å (2) 5400 Å (3) 4800 Å (4) 4000 Å

34. A power transmission line feeds input power at 2400 V to a step-down transformer with its primary windings having 6000 turns. The output power is delivered at 240 V by the transformer. If the current in the primary of the transformer is 6 A and its efficiency is 80%, the output current would be :

(1) 60 A (2) 54 A (3) 40 A (4) 48 A

35. An electron is travelling horizontally towards east. A magnetic field in vertically upward direction exerts a force on the electron along :

(1) west (2) east

(3) south (4) north

36. In fourth orbit of hydrogen atom, wavelength of revolving electron is λ. What is its circumference ?

(1) 8λ (2) 4λ

(3) λ (4) 2λ

37. A galvanometer has a coil of resistance 40 Ω and its full scale deflection current is 5 A. Value of shunt resistance that should be connected with it so that it can read a maximum of 85 A current, is :

(1) 2.5 Ω (2) 5 Ω

(3) 1.5 Ω (4) 0.75 Ω

38. A parallel plate capacitor is charged and the charging battery is then disconnected. If the plates of the capacitor are moved further apart by means of insulating handles, then which of the following statement is correct ?

(1) The charge on the capacitor increases.

(2) The capacitance increases.

(3) The voltage across the plates increases.

(4) The electrostatic energy stored in the capacitor decreases.

39. Fusion reaction takes place at high temperature as :

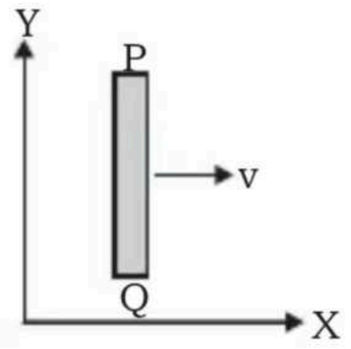
(1) atoms get ionised at high temperature

(2) kinetic energy is enough to overcome the coulomb repulsive force between nuclei at high temperature.

(3) nuclei break at high temperature.

(4) molecules break at higher temperature.

40. A conducting rod PQ moves parallel to X-axis in a uniform magnetic field, pointing in the positive Z-direction. The end P of the rod gets :



(1) first positively charged and then negatively charged

(2) neutral

(3) positively charged

(4) negatively charged

41. Which of the following statement is false ?

(1) In a balanced Wheatstone bridge if the cell and the galvanometer are exchanged, the null point is disturbed.

(2) Wheatstone bridge is the most sensitive when all the four resistances are of the same order of magnitude.

(3) Kirchhoff's second law represents energy conservation.

(4) A rheostat can be used as a potential divider.

42. A bomb travelling in a parabolic path under the effect of gravity, explodes in mid air. The centre of mass of fragments will :

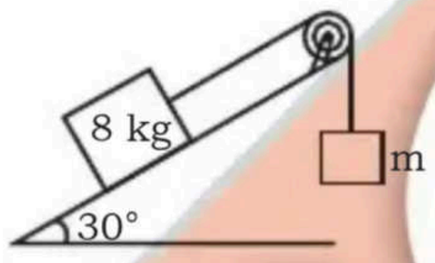
(1) move in irregular path

(2) move vertically upwards and then downwards

(3) move vertically downwards

(4) move in the parabolic path as unexploded bomb would have travelled

43. In Fig. below, coefficient of kinetic friction between the 8 kg block and the inclined surface is $\frac{1}{2\sqrt{3}}$. Here 'm' is such a mass that the 8 kg block is moving up the plane with a constant speed, then m is :



(1) 6 kg (2) 8 kg (3) 4 kg (4) $8\sqrt{3}$ kg

44. If wattless current flows in the AC circuit, then the circuit is :

(1) RC series circuit only

(2) LCR series circuit

(3) purely capacitive circuit

(4) purely resistive circuit

45. In case of projectile motion if two projectiles A and B are projected with same speed at angle 20° and 70° respectively to the horizontal, then:

I. $R_A < R_B$ II. $H_A < H_B$

III. $T_A > T_B$ IV. $T_A < T_B$

Choose the most appropriate option given below

(1) I, II, IV (2) II, IV (3) II, III (4) I, II, III

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46. Select the mismatch pair.

(1) Capsid — Protein coat of viroids.

(2) Low molecular wieght RNA — Viroid

(3) Abnormally folded protein — Prions

(4) Capsomeres — Sub unit of capsid

47. Match List I with List II.

List I	List II
A. Angina	I. Heart muscle is suddenly damaged.
B. Cardiac arrest	II. More common among the middle-aged and elderly.
C. Heart attack	III. Congestion of the lungs is one of the main symptoms.
D. Heart failure	IV. Heart stops beating.

Choose the correct answer from the options given below.

(1) (A-I), (B-IV), (C-III), (D-II)

(2) (A-II), (B-IV), (C-I), (D-III)

(3) (A-IV), (B-II), (C-I), (D-III)

(4) (A-II), (B-IV), (C-III), (D-I)

48. **Assertion(A):** The cells of testes have abundance of smooth endoplasmic reticulum.

Reason(R): The cells of testes secrete steroid/ lipid hormones.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are correct but Reason is not correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

49. In humans, male germ cells differentiate into _____ at the end of first meiotic division :

(1) Spermatozoa

(2) Spermatogonia

(3) Primary spermatocyte

(4) Secondary spermatocyte

50. Which of the following has same class but different order as compared to Leopard?

(1) Dog and Cat (2) Lion and Tiger

(3) Monkey and Human (4) Both (1) and (2)

51. How many bones given in the box are unpaired?

Lacrimal, Parietal, Ethmoid, Occipital, Mandible, Nasal, Hyoid, Scapula, Collar bone, Pubis.

(1) 5 (2) 4

(3) 3 (4) 6

52. Mark the incorrect statement with respect to cytokinesis.

(1) In animal cell appearance of furrow in the plasma membrane.

(2) In plant cell appearance of a cell plate.

(3) Both mitochondria and plastid get distributed between the daughter cells.

(4) Cytokinesis is not followed by karyokinesis as a result of which syncytium.

53. A epithelium which is composed of a single layer of tall and slender cells are found in the :

(1) Air sacs of lungs

(2) Ducts of glands

(3) Lining of stomach

(4) Moist surface of buccal cavity

54. Match the following column – I with column – II

	Column – I		Column – II
A.	NADP	i.	Zn
B.	Carboxy peptidase	ii.	Haem
C.	Pyruvate dehydrogenase	iii.	Coenzyme – A
D.	Peroxidase	iv.	Niacin
		v.	Mg

(1) A – iv, B – i, C – v, D – ii

(2) A – iv, B – i, C – iii,D – ii

(3) A – iii, B – ii,C – i, D – iv

(4) A – iv, B – i, C – iii, D – ii

55. The hormones that attain a peak level in the middle of menstrual cycle is/are :

A. FSH B. Thyroxine

C. Progesterone D. LH

Choose the correct option :

(1) A and D only (2) D only

(3) A, B, C and D (4) A only

56. Read the following statements and choose the correct option regarding sickle-cell anaemia.

(A) Shape of RBC from biconcave disc to enlongated sickle like structure.

(B) The disease is controlled by a single pair of allele.

(C) This occur due to single base substitution at the sixth codon of the beta globin chain.

(D) Homozyous individual for HbS (HbS HbS) show the disease phenotype.

(1) A and B (2) C and D

(3) A, B, C and D (4) All except D

57. Factors which can cause right shift in oxyhaemoglobin dissociation curve include all except :

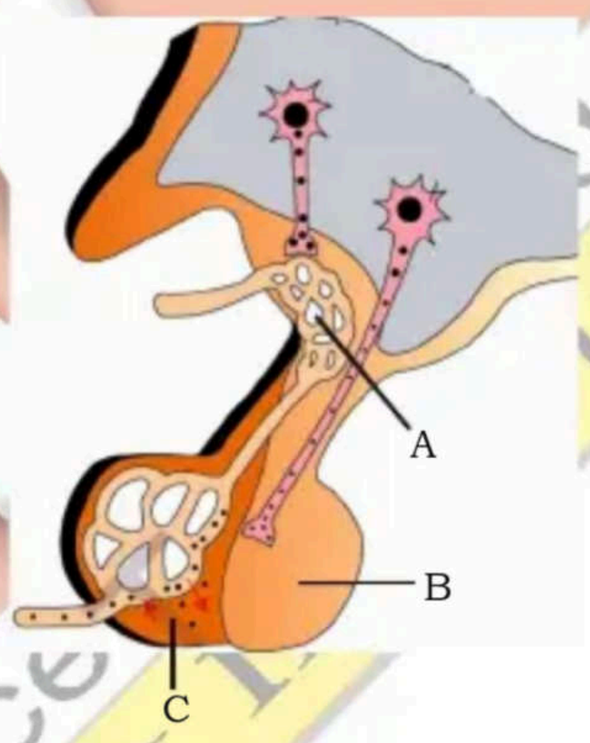
(1) increase in pO_2

(2) increase in temperature

(3) increase in pCO_2

(4) increase in hydrogen ion concentration.

58. In the following A, B and C labelling of pituitary given below. Which part stores and releases a hormone which can affect the kidney function by its constrictory effect on blood vessels ?



(1) B (2) C

(3) Both A and B (4) A

59. Which of the following is a characteristic of Mendelian inheritance?

(1) Idependent assortment of trait.

(2) Segregation of traits.

(3) Law of dominance.

(4) More than one option.

60. Seed coat is thin membranous and generally fixed with fruit wall.

(1) Gram and Maize (2) Ground nut and Gram

(3) Maize and Rice (4) Pea and Bean

61. **Statement – I:** In respiratory ETS-enegry of oxidation-reduction utilized for production of proton gradient.

Statement – II: In ETS of respiration oxidation of one carrier and reduction of another carrier is essential.

(1) Both Statements (I) and (II) are correct.

(2) Both Statements (I) and (II) are incorrect.

(3) Only Statement (I) is correct.

(4) Only Statement (II) is correct.

62. The secretions of the acrosome help the sperm to enter into the cytoplasm of ovum through :

(1) Cells of the corona radiata

(2) Uterus

(3) Zona pellucida

(4) Cervix

63. Mark the true statement with respect to telophase.

(1) Final stage of karyokinesis.

(2) Chromosome identity is lost as discrete elements.

(3) Chromosome cluster at opposite spindle.

(4) All of these.

64. **Assertion(A):** Meiosis, per-se results in reduction of chromosome number by half.

Reason(R): Meiosis conserves specific chromosome number of each species across generations in sexually reproducing organism.

(1) Both (A) and (R) are correct and (R) is correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) (R) is correct but (A) is incorrect.

65. In tubectomy procedure, a small part of the fallopian tube is :

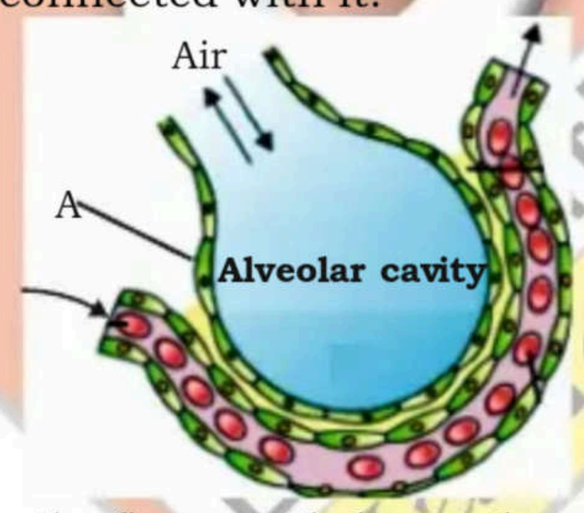
(1) tied up through a small incision in the abdomen

(2) removed

(3) tied up through vagina

(4) All of these

66. Identify the labelling 'A' and the nature of the tissue connected with it.



Observe the figure and choose the correct option.

(1) Red blood cell – Connective tissue.

(2) Alveolar wall – Simple squamous epithelium.

(3) Blood capillary – Endothelium.

(4) Basement substance – Squamous epithelium.

67. The stage between zygotene and diplotene is called:

(1) Leptotene (2) Interphase

(3) Interkinesis (4) Pachytene

68. Match List I with List II.

List I	**List II**
A. Aquatic arthropods	I. Pulmonary respiration
B. Insects	II. Exchange O_2 with CO_2 by simple diffusion
C. Reptiles	III. Tracheal respiration
D. Flatworm	IV. Branchial respiration

Choose the correct answer from the options given below.

(1) (A-III), (B-II), (C-I), (D-IV)

(2) (A-IV), (B-III), (C-I), (D-II)

(3) (A-IV), (B-II), (C-II), (D-I)

(4) (A-I), (B-III), (C-IV), (D-II)

69. What is the role of transcription factor in gene expression?

(1) Bind to RNA and enhance translation.

(2) Bind to DNA and enhance transcription.

(3) Bind to DNA and enhance replication.

(4) Bind to RNA and prevent replication.

70. Reabsorption of amino acids from the glomerular filtrate mainly takes place in the PCT by :

(1) Osmosis (2) Active transport

(3) Simple diffusion (4) Passive transport

71. The common point between perisperm and endosperm is/are:

(1) having reserve food

(2) both are tissues within a seed

(3) both are haploid

(4) both (1) and (2)

72. Insulin causes $\frac{\text{I}}{\text{____}}$ and stimulates $\frac{\text{II}}{\text{____}}$. Fill in the blanks and choose the correct option given below.

(1) I – Hyperglycemia, II – Glycogenolysis

(2) I – Hypoglycemia, II – Glycogenesis

(3) I –Hyperglycemia, II – Glycogenesis

(4) I – Hypoglycemia, II – Gluconeogenesis

73. Match List – I with List – II.

List – I	**List – II**
A. Palmitic acid	i. 2 – carbon atom
B. Arachidonic acid	ii. 16 – carbon atom
C. Glycerol	iii. 27 – carbon atom
D. Cholesterol	iv. 3 – carbon atom
E. Glycine	v. 20 – carbon atom

(1) A – ii, B – iii, C – iv, D – v, E – i

(2) A – iii, B – ii, C – iv, D – i, E – v

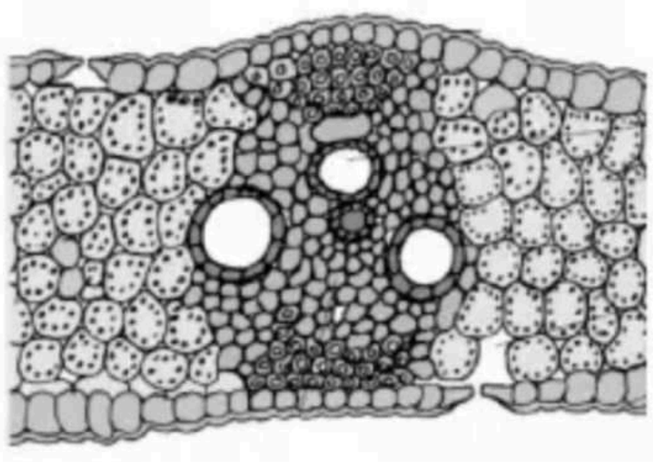
(3) A – ii, B – v, C – iv, D – iii, E – i

(4) A – iii, B – v, C – iv, D – ii, E – i

74. Introduction of fish like *Gambusia* in ponds and spraying of insecticides in swamps can result in decline in spread of all disease except :

(1) Chikungunya (2) Malaria

(3) Dengue (4) Pneumonia

75. 

Select the incorrect option with respect to above diagram.

(1) The stomata are present on both the surface of the epidermis.

(2) Mesophyll is not differentiated into palisade and spongy parenchyma.

(3) Similar sizes of vascular bundles (except in main vain)

(4) The vein vary in thickness in the parallel venation of

76. Inner parts of cerebral hemispheres and a group of associated deep structures forms a complex structure which is concerned with :

(1) olfaction.

(2) regulation of sexual behaviour.

(3) autonomic responses.

(4) All of these.

77. Mark the incorrect match pair.

(1) Adventitious roots → Monstera, Banyan

(2) Pneumatophore → Rhizophora

(3) Tap root → Mustard and Sugarcane

(4) Stilt root → Maize and Sugarcane

78. Which of the following is correct sequence with respect to embryonic stages in humans ?

(1) Cleavage → morula → gastrula → blastula → embryo

(2) Morula → blastomeres → blastocyst → cleavage → embryo

(3) Cleavage → morula → blastula → gastrula → embryo

(4) Morula → blastocyst → blastomeres → cleavage

79. Mark the incorrect option with respect to mycoplasma.

(1) Wall–less prokaryote

(2) Pathogentic to both animals and plant

(3) Smallest monera

(4) They cannot survive without oxygen

80. Which of the following classification system organism belonging to the same taxa have a common ancestor?

(1) Natural system of classification

(2) Aritificial system of classification

(3) Cytotaxonomy

(4) Phylogenetic system of classification

81. Consider the following characteristics.

A. Air bladder is present which regulates buoyancy.

B. Sexes are seperate.

C. In males pelvic fins bear claspers.

Select the correct option for the above given characteristics.

(1) A and C are characteristics of Angel fish.

(2) B and C are characteristics of Sea horse.

(3) A and B are characteristics of *Clarias*.

(4) A and B are characteristics of Dog fish.

82. Kreb's cycle start with condesation of ___(i)__ with __(ii)__ and H_2O to yield citric acid. Select the correct option to fill in the blanks (i) and (ii).

(1) i – oxaloacetic acid, ii – acetyl group

(2) i – pyruvie acid, ii – malic acid

(3) i – oxaloacetic acid, ii – citric acid

(4) i – oxaloacetic acid, ii – pyruvic acid

83. Which part of respiratory tract is not supported by incomplete cartilaginous rings ?

(1) Secondary bronchi (2) Initial bronchioles

(3) Tertiary bronchi (4) Terminal bronchioles

84. Which of the following is not a heredity disease?

(1) Colourblindness (2) Sickle-cell anaemia

(3) *Phenylketonuria* (4) HIV/AIDS

85. Given below are two statements :

Statement I : Transmission of an impulse across chemical synapses is very similar to impulse conduction along a single axon.

Statement II : A nerve impulse is transmitted from one neuron to another through junction called neurotransmitters.

In the light of above statements, choose the correct answer from the options given below.

(1) Only **Statement I** is correct.

(2) Both **Statement I** and **Statement II** are correct.

(3) Only **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are incorrect.

86. Mark the incorrect pair.

(1) ⊕♂♀ $k_{2+2}\ C_4\ A_{2+4}\ \underline{G}_{(2)}$ - Cabbage, Radish, Cauliflower

(2) Br %♂♀ $P_2\ A_3\ \underline{G}_1$ (or) 3 - Sugarcane, Maize, doob, grass, Wheat

(3) Br %♀ k papuus $C_{(5)}A_0\ \overline{G}_{(2)}$ - Marigold, Sunflower, Lettuce

(4) Br/EBr⊕♂♀ Epi 3-7 $k_{(5)}\ C_5A \infty \underline{G}(5-\infty)$ - Cotton, Okra, Dahlia

87. Which of the following vector is normally used in gene therapy for humans in severe combined immunodeficiency disease ?

(1) Agrobacterium tumifaciens

(2) Plasmid

(3) Bacteriophage

(4) Disarmed retrovirus

88. The post–transcriptional modification, capping, splicing and tailing is required:

(1) for the primary transcript in eukaryotes to make it functional.

(2) for all types of RNAs in bacteria.

(3) for incorporating promotor and terminator in mRNA in prokaryotes.

(4) All of these

89. Antibiotics have greatly improved our capacity to treat deadly disease such as :

(1) Tetanus, Plague, Cancer.

(2) Diphtheria, Leprosy, Plague.

(3) Whooping cough, Malaria, Dengue.

(4) Chikungunya, Leprosy, Plague.

90. Which of the following is not a feature of meristematic cells?

(1) Presence of few of plasomdesmata.

(2) Presence of cell wall that is capable of growth.

(3) Small size & dense protoplasm

(4) Capability to undergo cytokinesis and karyokinesis.

91. Arrange the given steps of recombinant DNA technology in a proper sequence.

A. Extraction of the desired product.

B. Construction of a recombinant DNA molecule.

C. Multiplication of r-DNA into the host cell.

D. Introduction of r-DNA into the host cell.

Choose the correct sequence.

(1) C → A → D → E (2) B → D → C → A

(3) B → A → C → D (4) A → B → D → C

92. Which component of xylem tissue which are elongated (or) tube like cells, thick and lignified wall and tapering ends?

(1) Xylem fibres (2) Xylem parenchyma

(3) Xylem vessel (4) Xylem trachieds

93. Which of the following statement is incorrect ?

(1) Comparative anatomy and morphology shows similarities and differences among organisms of today and those that existed years ago.

(2) Sweet potato (root modification) and potato (stem modification) is an example of analogy.

(3) The first non-cellular forms of life could have originated 3 billion years back.

(4) Jawless fish probably evolved around 320 mya.

94. Match List – I and List – II.

List – I	**List – II**
A. Chromatin	I. T.W Engelmann
B. TCA cycle	II. Conelius Van Niel
C. Action specturm	III. Hans Krebs
D. Microbiologist	IV. Flemming

(1) A – II, B – I, C – III, D – IV

(2) A – IV, B – III, C – I, D – II

(3) A – III, B – IV, C – II, D – I

(4) A – IV, B – III, C – II, D – I

95. Which of the following enzyme enclosed with HIV which produces DNA copy of the viral genome after entry of virus into the host cells ?

(1) Carbonic anhydrase

(2) DNA polymerase

(3) Reverse transcriptase

(4) RNA polymerase

96. Select the mismatch pair.

(1) Taxon – A taxonomic group of any rank.

(2) Species – Basic unit of classification.

(3) Phylum – Division is the same category in case of plants.

(4) Order – Assemble of class with few similar characters.

97. Given below are two statements – one is labelled as Assertion (A) and other as Reason (R).

Assertion (A) : Different varieties of cheese are known by their characteristic texture, flavour and taste, the specificity coming from the microbes used.

Reason (R) : Large holes in 'Swiss cheese' are due to production of a large amount of CO_2 by a fungi named *Propionibacterium sharmanii*.

In the light of above statements, choose the correct answer from the options given below.

(1) **(A)** is true but **(R)** is false.

(2) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.

(3) Both **(A)** and **(R)** are false.

(4) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

98. Which of the following plant is characterized with the presence of reticulate venation in leaves and endospermic seed ?

(1) Gram and Petunia

(2) Bean and Pea

(3) Castor and Petunia

(4) Mustard and Orchid

99. How many of the following statement is/are not incorrect ?

A. In reptiles, the left atrium receives oxygenated blood from the lungs.

B. Ventricular systole increases the ventricular pressure causing the closure of semilunar valves.

C. During a cardiac cycle, each ventricle pumps out approximately 70 mL of blood which is called the beat volume.

D. Adrenal medullary hormones decrease the cardiac output.

(1) 2 (2) 3

(3) 1 (4) 4

100. Cell division and xylem differentiation is mainly controlled by

(1) Ethylene (2) Cytokinin

(3) ABA (4) Auxin

101. Some of the salient observation drawn from human genome project are as follow:

(A) Less than 2% of genome codes for proteins.

(B) SNPs pronounced as 'snips'

(C) Chromosome Y has the fewest (231) gene.

(D) Largest known human gene being dystrophin at 2.4 million bases.

(E) Human genome contian 3164.7 bp.

Choose the correct option:

(1) All except B (2) All except E

(3) A, B, C and E (4) All of these

102. Meiosis take place in

(1) meiocyte

(2) megaspore mother cell and microspore mother cell

(3) Some cells of sporophyte in bryophyte

(4) All of these

103. Each ovary of *Periplaneta americana* is formed of a group of :

(1) 8 ovarioles

(2) 14–16 ovarian tubules

(3) 9–10 ovarioles

(4) 16 ovarian tubules

104. Lysosomes are:

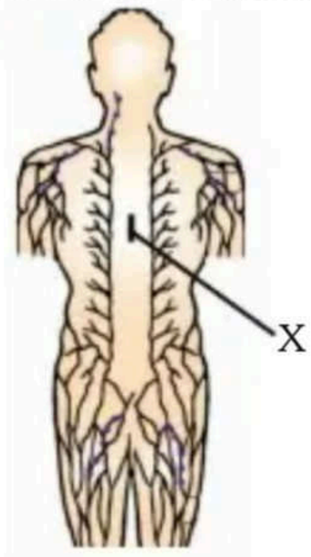
(1) not a part of endomembrane system.

(2) membrane less cell organelle.

(3) contain hydrolytic enzyme which are active in acidic pH.

(4) formed by the process of packaging in SER.

105. Choose the correct option with respect to identification and function of labelled structure 'X' in the diagrammatic representation of lymphatic system of human body given below.



(1) Spleen — Large reservior of erythrocytes.

(2) Tonsils — Located near the heart and beneath the breast bone.

(3) Thymus — Development and maturation of T-lymphocytes.

(4) Lymph nodes — Trapped antigens and cause the immune response.

106. A man with 47 chromosome due to three copied of chromosome 21 is characterized by:

(1) Trisomy of 21 (2) Triploidy

(3) Down's syndrome (4) Both (1) and (3)

107. Given below are two statements.

Statement I : The separated DNA fragments can be visualised only after staining the DNA with ethidium bromide exposed to gamma radiation.

Statement II : Plasmids and bacteriophages have the ability to replicate within bacterial cells independent of the control of chromosomal DNA.

In the light of above statements, choose the correct answer from the options given below.

(1) Only **Statement II** is incorrect.

(2) Both **Statement I** and **Statement II** are correct.

(3) Only **Statement I** is incorrect.

(4) Both **Statement I** and **Statement II** are incorrect.

108. The excretory system of *Rana tigrina* consists of a pair of :

(1) Kidneys, urethra, cloaca and urinary bladder.

(2) Cloaca, ureters, urethra and urinary bladder.

(3) Ureters, cloaca, kidneys and urinary bladder.

(4) Urethra, cloaca and urinary bladder only.

109. Flagellar movement helps in the :

(1) Swimming of spermatozoa.

(2) Maintenance of water current in the canal system of sponges.

(3) Locomotion of *Euglena*.

(4) All of these

110. Match List – I with List – II

	List – I		List – II
A.	Repetitive DNA	i.	1.4 million location where single base DNA difference.
B.	Polymorphism	ii.	Coding sequence of RNA expressed.
C.	SNPs	iii.	U.S. Department of energy.
D.	HGP	iv.	Arises due to mutation.
E.	Express sequence tags	v.	Radio active DNA as a probe.
		vi.	Identify difference in some specific region in DNA sequence.

(1) A – ii, B – i, C – iv, D – vi, E – v

(2) A – vi, B – iv, C – i, D – iii, E –ii

(3) A – iv, B – i, C – vi, D – v, E – iii

(4) A – ii, B – iv, C – iii, D – i, E – v

111. Select the mismatch pair.

Excretory structures	Organisms
(1) Nephridia	Earthworms
(2) Green glands	Prawns
(3) Flame cells	*Planaria*
(4) Protonephridia	Roundworm

112. All of the following are asexual spores in fungi except:

(1) Zoospore

(2) Sporangiospore

(3) Conidia

(4) Ascospore and oospores

113. Arborescent lycopods evolved from :

(1) Zosterophyllum (2) Herbaceous lycopods

(3) Progymnosperms (4) Psilophyton

114. **Statement – I:** The process of splicing represents the dominance of RNA world.

Statement – II: The split gene arrangement represent probably the ancient feature of genome.

(1) Both statements (I) and (II) are correct.

(2) Both statements (I) and (II) are incorrect.

(3) Only statement (I) is correct.

(4) Only statement (II) is correct.

115. Digestive system is complete with mouth on the lower (ventral) side and anus on the upper (dorsal) side in :

(1) *Ophiura* (2) *Physalia*

(3) *Aplysia* (4) *Saccoglossus*

116. How many of the following are broadly utilitarian services of the ecosystem?

Fibre, Firewood, Lubricants, Resins, Cereals, Oxygen , Pollination

(1) Three (2) Two

(3) Four (4) Five

117. Which of the following hormone induces ovulation of graafian follicles and maintains the corpus luteum, formed from the remnants of the graffian follicles after ovulation ?

(1) Thyroid stimulating hormone

(2) Follicle stimulating hormone

(3) Luteinizing hormone

(4) Progesterone

118. The capacity to generate a whole plant from any cell/explant and method of producing thousands of plants through tissue culture is called respectively :

(1) Somatic hybrids and Totipotency.

(2) Totipotency and Micro-propagation.

(3) Micro-propagation and Totipotency.

(4) Totipotency and Somatic hybridisation.

119. Find the incorrect pair.

(1) Hypocotyl — Terminate with radicle

(2) Micropyle — Present in seed coat

(3) Radicle — Enclosed by coleorrhiza

(4) Coleoptile — Inner covering of endosperm.

120. The function of all the following is to maintain the ionic balance and pH of the body fluids by selective secretion of hydrogen ions except.

(1) Distal convoluted tubule

(2) Descending limb of loop of Henle

(3) Collecting duct

(4) Proximal convoluted tubule

121. Which among the following is living part of mechanical tissue?

(1) Sclerenchyma and Collenchyma

(2) Phloem fibres

(3) Xylem and Phloem parenchyma

(4) Both (1) and (2)

122. Recognition site for which of the following restriction enzyme is present in amp^R gene of cloning vector pBR322 ?

(1) *Cla* I (2) *Pvu* II (3) *Sal* I (4) *Pvu* I

123. Which of the following statement are incorrect?

(1) Collagen act as intercellular ground substance.

(2) Antibody acts as fight infectious agents.

(3) GLUT – 4 enables glucose transport into cells.

(4) Trypsin act as intercellular messengers.

124. Isolation of genetic material from bacterial cells involves the use of :

(1) Cellulase (2) Chitinase

(3) Lysozyme (4) Ribonuclease

125. Third tropic level in ecosystem can be represented by:

(1) Wolf, small fish and lion

(2) Zooplankton, grasshopper and cow

(3) Man and lion

(4) Phytoplankton, trees and grass

126. If a female menstrual cycle is of 32 days, then the fertile period in which a couple should avoid coitus is from day :

(1) 14 – 21 (2) 32 – 39 (3) 16 – 27 (4) 22 – 29

127. Epigynous flowers are found in:

(1) cucumber and guava

(2) peach and rose

(3) ray floret of sun flower

(4) both (1) and (3)

128. Consider the following tropic level.

(i) Tree → Birds → Parasites

(ii) Tree → Insect → Insect eating bird

(iii) Grasses → Cow → Lion

(iv) Phytoplanktons → Zooplankton → Small fish → Large fish

(v) Deteritus → Detritivores → Worm eating bird

How many tropic level represent upright pyramid of number in a GFC?

(1) Two (2) Three

(3) Four (4) One

129. When a skeletal muscle contracts :

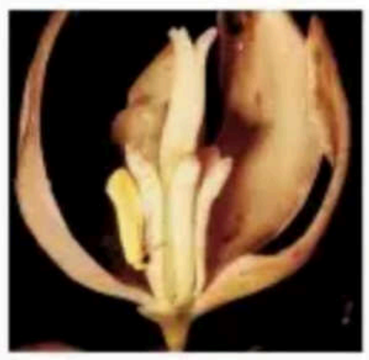
(1) 'I' bands remains unaffected.

(2) H-zone increases in length.

(3) 'I' bands get reduced.

(4) 'A' bands increases in length.

130. The following figure shows:



(1) Parasitism (2) Mutualism

(3) Amensalism (4) Commensalism

131. Biopiracy is the term used to refer to the use of bio-resources by multinational companies and other organisation :

(1) with proper authorisation from the countries and people concerned without compensatory payment.

(2) without proper authorisation from the countries and people concerned without compensatory payment.

(3) got patent rights through US patent and trademark office.

(4) Both '2' and '3'

132. The process of mineralisation by microorganisms help in the release of

(1) organic nutrient from humus.

(2) inorganic nutrient from humus.

(3) inorganic nutrients from detritus and formation of humus.

(4) both organic and inorganic nutrient from detritus.

133. Which among the following is not a common function of stem in majority of Angiosperms ?

(1) Spreading out branches bearing leaves, flower and fruit.

(2) It conduct water, minerals and photosynthesis.

(3) Absorption of water and minerals.

(4) To store food, support, protection and of vegetative propagation.

134. Which of the following can exhibit radial or bilateral symmetry depeneding on their stage?

(1) Sea-fan (2) Sea lily

(3) Sea anemone (4) *Obelia*

135. The patients are given substances called biological response modifiers such as α-interferon which activates their immune system and help in treatment of :

(1) Polio (2) Small pox

(3) Cancer (4) Pneumonia

CHEMISTRY

136. A solution contains I_2 in benzene. The mole fraction of I_2 is 0.2. Calculate molarity of solution if density of solution is d g/mL.

(Molecular weight of I_2 = 254 g/mol).

(1) 1.77 d (2) 17.7 d

(3) 1.17 d (4) 2.77 d

137. For a first order reaction : A → P, the temperature (T) dependent rate constant (K) was found to follow the equation $\log K = -(2000)\frac{1}{T} + 6.0$. The pre-exponential factor (A) and the activation energy (E_a) respectively are :

(1) $1.0 \times 10^6 s^{-1}$ and 9.2 kJ mol^{-1}

(2) 6.0 s^{-1} and 16.6 kJ mol^{-1}

(3) $1.0 \times 10^6 s^{-1}$ and 16.6 kJ mol^{-1}

(4) $1.0 \times 10^6 s^{-1}$ and 38.3 kJ mol^{-1}

138. Match the Column-I with Column-II.

Column I (Classification of molecules based on octet rule)	**Column II (Example)**
A. Molecules follow octet rule	I. NO
B. Molecules with incomplete octet	II. BCl_3
C. Molecules with incomplete octet with odd electron	III. H_2SO_4
D. Molecules with expanded octet	IV. CO_2

(1) (A-IV), (B-II), (C-I), (D-III)

(2) (A-I), (B-II), (C-III), (D-IV)

(3) (A-II), (B-I), (C-IV), (D-III)

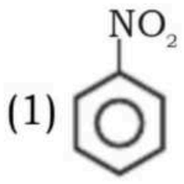
(4) (A-IV), (B-III), (C-II), (D-I)

139. Which one of the follolwing compound contains α-1-4 glycosidic linkage ?

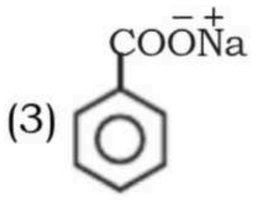
(1) Sucrose (2) Maltose

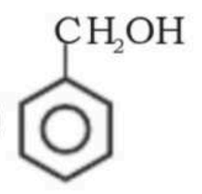
(3) Lactose (4) Cellulose

140. 

(1) 

(2) 

(3) 

(4) 

141. The d-electronic configuration of an octahedral Co(II) complex having magnetic moment of 3.95 BM is :

(1) $t_{2g}^6 e_g^1$ (2) $t_{2g}^3 e_g^0$

(3) $t_{2g}^5 e_g^2$ (4) $e_g^4 t_{2g}^3$

142. **Assertion (A) :** Experimentally determined oxygen-oxygen bond lengths in O_3 are found to be same and the bond length is greater than that of a O==O (double bond) but less than that of a single (O—O) bond.

Reason (R) : The strong lone pair-lone pair repulsion between oxygen atom is solely responsible for the fact that the bond length in ozone is smaller than that of a double bond (O=O) but more than that of a single bond (O—O).

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is not the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

143. If molar mass of HCl is M then it's equivalent mass in the redox reaction :

$KMnO_4 + H_2C_2O_4 + HCl \longrightarrow$

$MnCl_2 + CO_2 + 2KCl$ is given by :

(1) $\frac{3M}{5}$ (2) $\frac{M}{3}$

(3) $\frac{M}{2}$ (4) M

144. The oxidation potential of 0.1 M solution of M^{+2} ions is if $E^0_{M^{+2}/M} = -1.18V$:

(1) – 1.15 V (2) 1.15 V

(3) – 1.21 V (4) 1.21 V

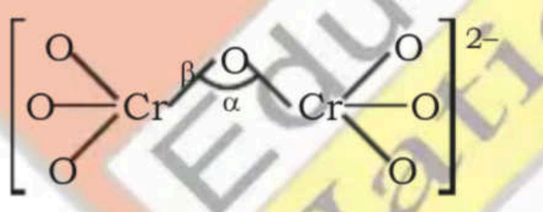
145. The number of element from the following that do not belong to actinoids is :

Er, Cm, Ho, Md, No, Tm and Dy

(1) 4 (2) 6

(3) 5 (4) 3

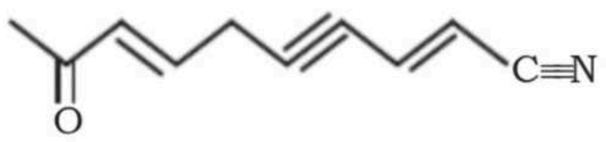
146. In given pictorial view of dichromate ion, the 'α' and 'β' is, respectively were α is in degree and β is in Picometer (Pm).



(1) 123°, 173 Pm (2) 128°, 169 Pm

(3) 126°, 179 Pm (4) 136°, 179Pm

147. In the given structure number of sp and sp^2 hybridized carbon atoms present respectively are :



(1) 3 and 6 (2) 3 and 5

(3) 4 and 6 (4) 4 and 5

148. Which one of the following carbocation from the following is most stable ?

(1) $\overset{\oplus}{C}H_2 - CH = CH - CH_2 - OCH_3$

(2) $\overset{\oplus}{C}H_2 - CH = CH - OCH_3$

(3) $\overset{\oplus}{C}H_2 - CH = CH - OCOCH_3$

(4) $\overset{\oplus}{C}H_2 - CH = CH - F$

149. Which of the following is correct for a non-ideal solution ?

(1) $\Delta H_{mix} = 0$ (2) $\Delta V_{mix} = 0$

(3) $P_A = P_A^0 X_A$ (4) $\Delta S_{mix} > 0$

150. Which of the following aqueous solution will have highest boiling point ?

(1) 0.1 m glucose (2) 0.1 m K_2SO_4

(3) 0.1 m NaCl (4) 0.01 m $K_4[Fe(CN)_6]$

151. In which of the following arrangements, the sequence is not according to the property written against it :

(1) O > S > Se > Te : Boiling point

(2) HI > HBr > HCl > HF : Acidic strength

(3) $Cl_2 > Br_2 > F_2 > I_2$: Bond dissociation enthalpy

(4) N > O > C > B : First ionisation enthalpy

152. Which of the following ion that are expected to be behave as oxidising agent is ?

(1) Tl^+ (2) Tl^{3+}

(3) Pb^{2+} (4) Both '1' and '2'

153. Given below are two statements :

Statement I : Number of reacting species taking part in an elementary reaction which must collide simultaneously in order to bring about a chemical reaction is called molecularity of a reaction.

Statement II : Order of reaction is applicable to elementary reaction only, and for complex reaction order has no meaning.

In light of above statements, choose the correct answer from the options given below :

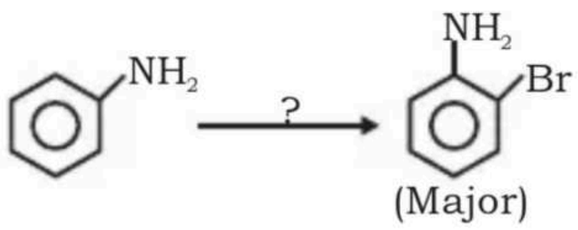
(1) Both **Statement I** and **Statement II** are true.

(2) Both **Statement I** and **Statement II** are false.

(3) **Statement I** is true but **Statement II** is false.

(4) **Statement I** is false but **Statement II** is true.

154. For the given reaction select the correct set of reagents is :



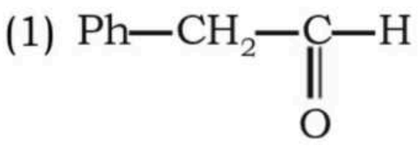
(1) $Br_2/FeBr_3, H_2O(\Delta), NaOH$

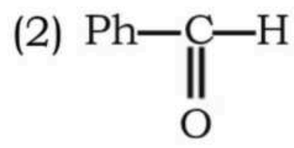
(2) $H_2SO_4, Ac_2O, Br_2, H_2O(\Delta), NaOH$

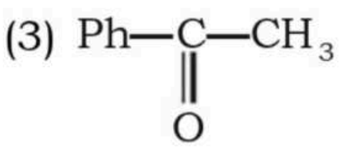
(3) $Ac_2O, Br_2, H_2O(\Delta), NaOH$

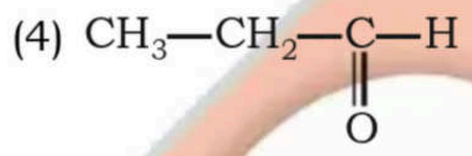
(4) $Ac_2O, H_2SO_4, Br_2, NaOH$

155. Which of the following compound given orange ppt with 2,4-DNP but do not give silver mirror with Tollen's reagent.

(1) 

(2) 

(3) 

(4) 

156. Enthalpy of neutralization of which of the following acid, base is maximum ?

(1) HCN and NaOH (2) HCl and KOH

(3) HCl and NH_4OH (4) HCOOH and NaOH

157. Equivalent conductivity of $Fe_2(SO_4)_3$ is related to molar conductivity by the expression :

(1) $\Lambda_{eq} = \Lambda_m$ (2) $\Lambda_{eq} = \frac{\Lambda_m}{3}$

(3) $\Lambda_{eq} = 3\Lambda_m$ (4) $\Lambda_{eq} = \frac{\Lambda_m}{6}$

158. Which of the ions from the following liberate hydrogen from a dilute acid is ?

(1) V^{2+} (2) Co^{3+}

(3) Cr^{3+} (4) Fe^{+2}

159. The product obtained by reductive ozonolysis of 1-phenylbutene is

(1) Propanal, Benzaldehyde

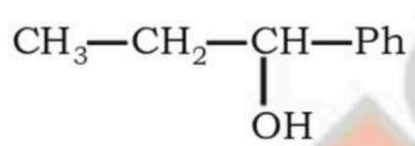
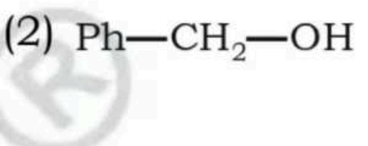
(2) Propanoic acid, Benzoic acid

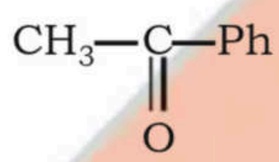
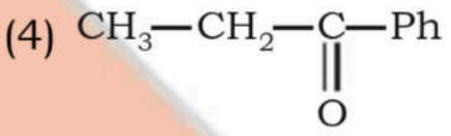
(3) Propanal, Benzoic acid

(4) Propanoic acid, Benzaldehyde

160. A $\xrightarrow{I_2/Ca(OH)_2} CHI_3 + (PhCOO)_2Ca$

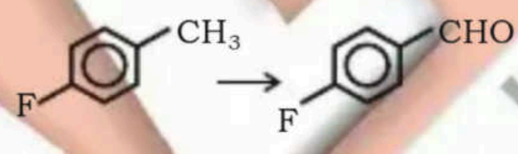
the compound 'A' is

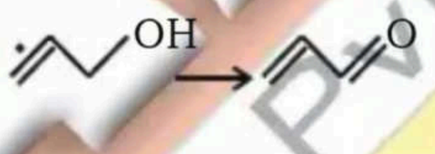
(1)  (2) 

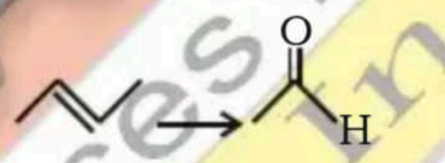
(3)  (4) 

161. Match the Column-I with Column-II

A. $CH_3CN \longrightarrow CH_3CHO$ i. P.C.C.

B.  ii. $CrO_2Cl_2;H_2O$

C.  iii. O_3/H_2O–Zn

D.  iv. DiBAL–H, $H_3O^{\oplus}$

(1) A → i, B → ii, C → iii, D → iv

(2) A → iii, B → ii, C → i, D → iv

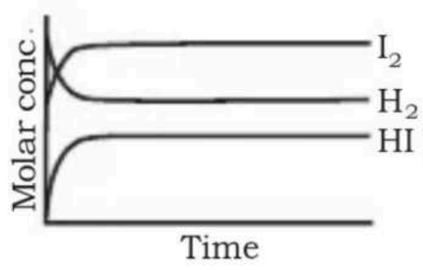
(3) A → iv, B → ii, C → i, D → iii

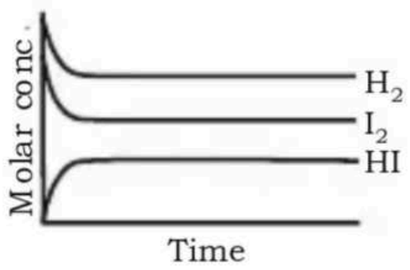
(4) A → i, B → iv, C → iii, D → ii

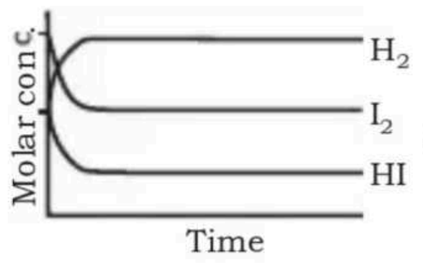
162. For the reaction,

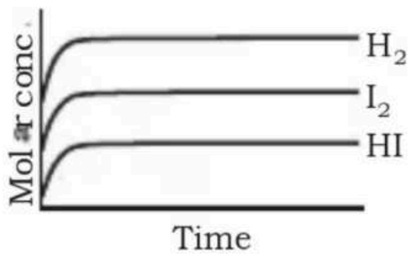
$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

Attainment of equilibrium is predicted correctly by :

(1) 

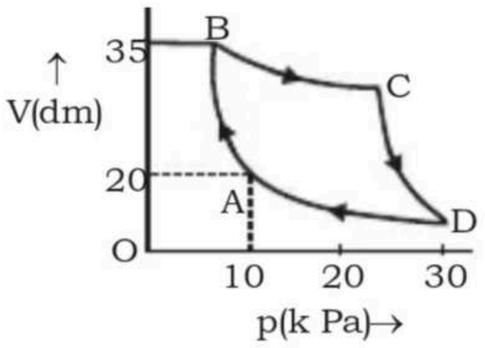
(2) 

(3) 

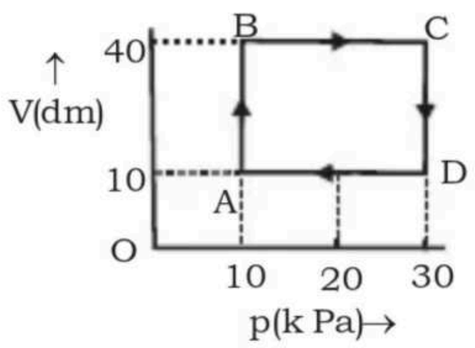
(4) 

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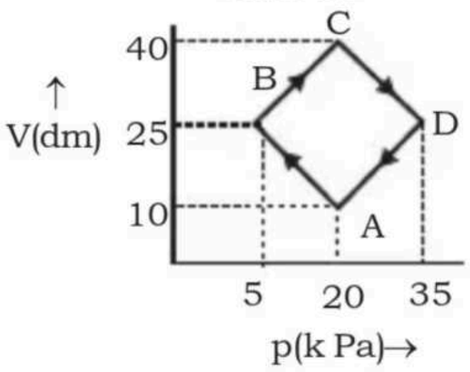
163. Case-I



Case-II



Case-III



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ as shown in the three cases above.

Choose the correct option regarding ΔU .

(1) ΔU (Case-III) > ΔU (Case-II) > ΔU (Case-I)

(2) ΔU (Case-I) > ΔU (Case-II) > ΔU (Case-III)

(3) ΔU (Case-I) > ΔU (Case-III) > ΔU (Case-II)

(4) ΔU (Case-I) = ΔU (Case-II) = ΔU (Case-III)

164. Heat treatment of muscular pain involves radiation of wavelength of about 900 nm. Which spectral line of H atom is suitable for this ?

Given : Rydberg constant

$R_H = 10^5\ cm^{-1}$, $h = 6.6 \times 10^{-34}$ J s, $c = 3 \times 10^8$ m/s)

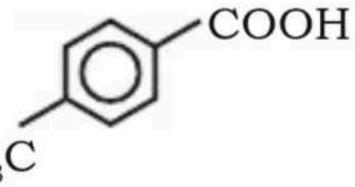
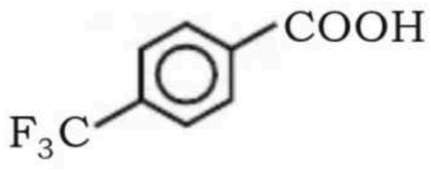
(1) Paschen series, $\infty \rightarrow 3$

(2) Lyman series, $\infty \rightarrow 1$

(3) Balmer series, $\infty \rightarrow 2$

(4) Paschen series, $5 \rightarrow 3$

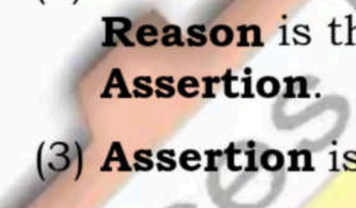
165. In which of the following I is more acidic than II.

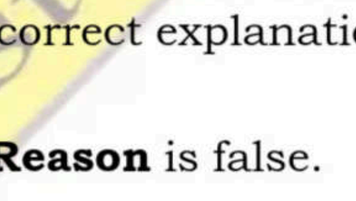
	I	II
(1)	CH_3CO_2H	CH_2FCO_2H
(2)	CH_2FCO_2H	CH_2ClCO_2H
(3)	$CH_3CHFCH_2CO_2H$	$CH_2FCH_2CO_2H$
(4)		

166. Ethyl-3-oxobutanoate

$$\xrightarrow[(ii)H^{\oplus}]{(i)NaBH_4} (X) \xrightarrow{CH_3CH_2-MgBr} \underset{\text{(Major)}}{(Y)}$$

the 'Y' is

(1) 

(2) 

(3) 

(4) $CH_3 - CH_3$

167. **Assertion:** Addition of HBr to propene Yields 2-Bromopropane but in presence of a peroxide it yield 1-bromopropane.

Reason: The reaction in presence of peroxide it follows free radical mechanism and 2° free radical is more stable than 1° free radical.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is the not the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

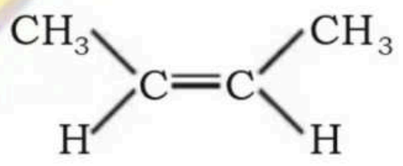
(4) **Assertion** is false but **Reason** is true.

168. The complex that can show facial and meridonial isomerism is :

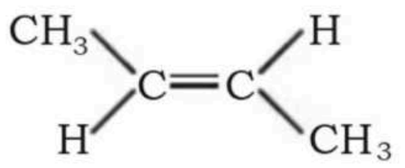
(1) $[Co(NH_3)_3(NO_2)_3]$ (2) $[Pt(NH_3)_2Cl_2]$

(3) $[Co(NH_3)_4Cl_2]^+$ (4) $[CoCl_2(en)_2]$

169. Which of the following statement(s) is correct regarding following structure ?



A



B

(1) A and B are stereoisomers of each other.

(2) Boiling point of A is more than B.

(3) A has more dipole moment than B.

(4) All of these

170. The incorrect matching pair is

	Column-I [Elements]	**Column-II [PPt color in Lassiagne test]**
(1)	Nitrogen	→ Prussian Blue
(2)	Sulphur	→ Violet
(3)	Chlorine	→ Blue
(4)	Phosphorus	→ Yellow

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171. 2.9×10^{-3} mol of CO_2 is left after removing 10^{21} molecules from its 'x' mg sample. The mass of CO_2 taken initially is

Given : $N_A = 6.02 \times 10^{23}$ mol^{-1}

(1) 200.6 mg (2) 98.3 mg

(3) 150.4 mg (4) 48.2 mg

172. Given below are two statements :

Statement I : For a given shell, the total number of allowed orbitals is given by n^2.

Statement II : For any subshell, the spatial orientation of the orbitals is given by $-l$ to $+l$ values including zero.

In the light of the above statements, choose the correct answer from the options given below :

(1) **Statement I** is true but **Statement II** is false.

(2) **Statement I** is false but **Statement II** is true.

(3) Both **Statement I** and **Statement II** are true.

(4) Both **Statement I** and **Statement II** are false.

173. The effect of temperature on spontaneity of reactions are represented as :

	ΔH	**ΔS**	**Temperature**	**Spontaneity**
(A)	+	–	any T	Non spontaneous
(B)	+	+	$T < T_e$	Spontaneous
(C)	–	–	$T < T_e$	Nonspontaneous
(D)	–	+	any T	spontaneous

Select correct ones :

(1) (B) and (D) only (2) (A) and (D) only

(3) (B) and (C) only (4) (A) and (C) only

174. Which amongst the following reaction of alkyl halide produces nitroalkane as a major product.

A. $R-X + KNO_2 \rightarrow$ B. $R-X + AgNO_2 \rightarrow$

C. $R-X + R'NH_2 \rightarrow$ D. $R-X + Et-NH-Me \rightarrow$

Select the most appropriate answer from the following option given below.

(1) A only (2) A and C only

(3) B and D only (4) B only

175. Which of the following statements are correct regarding Kolbe's electrolytic method ?

A. Alkanes containing even number of C-atoms are prepared.

B. Potasium salt of a carboxylic acid on electrolysis gives alkane.

C. Ethane cannot be prepared by this method.

D. H_2 gas is liberated at anode.

Choose the correct statement.

(1) A, B (2) B, C

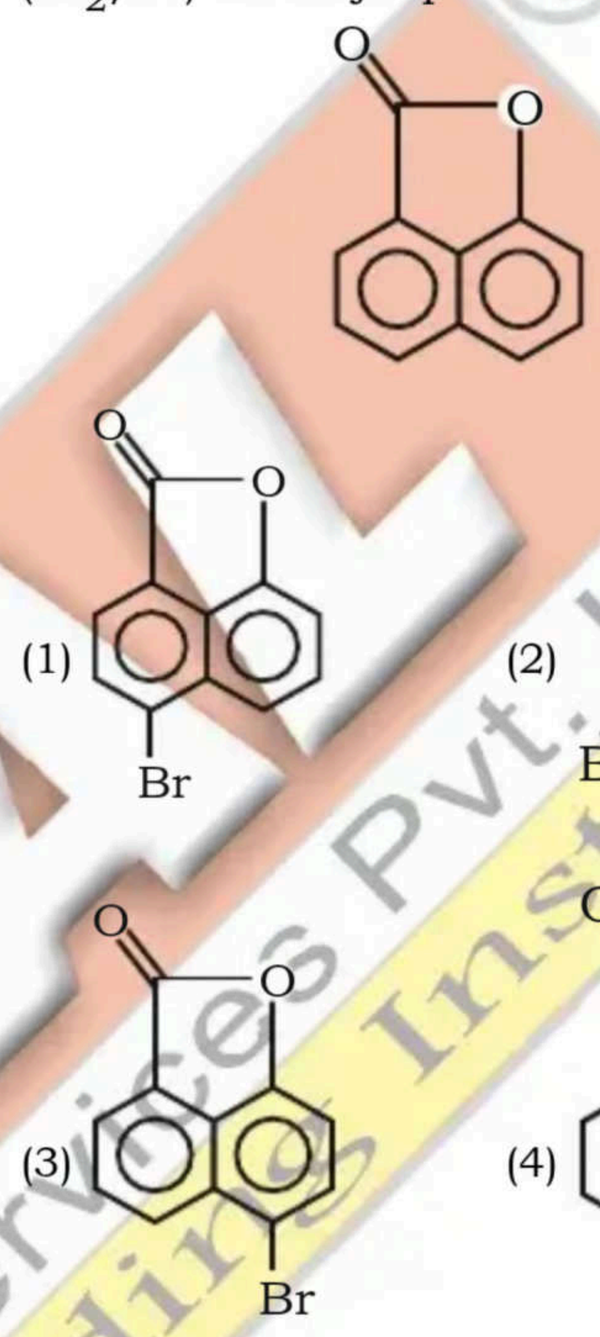
(3) A, C (4) A,B,C

176. Calculate the enthalpy change when 1 mole of water at 27°C and 1 atm pressure is converted to ice at 0°C. Given the enthalpy of fusion of ice is 5.96 kJ mol^{-1} and heat capacity of water is 4.2 J°c^{-1}g^{-1}.

(1) – 8.00 kJ mol^{-1} (2) 6.36 kJ mol^{-1}

(3) – 5.96 kJ mol^{-1} (4) 5.96 kJ mol^{-1}

177. If the given compound is subject to halogenation (Br_2/Fe) the major product formed is



178. Match the Column-I with Column-II.

Column I	**Column II**
A. Cl_2O_7	I. Amphoteric
B. PbO_2	II. Basic
C. Na_2O	III. Neutral
D. NO	IV. Acidic

(1) (A-I), (B-II), (C-III), (D-IV)

(2) (A-III), (B-I), (C-IV), (D-II)

(3) (A-IV), (B-I), (C-II), (D-III)

(4) (A-IV), (B-III), (C-II), (D-I)

179. Which of the following are not an isostructural pair ?

(1) XeF_2, IF_2^- (2) SiF_4, PCl_4^+

(3) SO_4^{2-}, CrO_4^{2-} (4) NH_3, NO_3^-

180. The electronic configuration of Curium (Z = 96) is :

(1) $[Rn]5f^7\,6d^1\,7s^2$ (2) $[Rn]5f^6\,6d^2\,7s^2$

(3) $[Rn]5f^9\,6d^1$ (4) $[Rn]5f^{10}$

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PHYSICS

02. $mg \sin\theta = \mu_k mg \cos\theta$

$\Rightarrow \tan\theta = \mu_k$

03. $y = A\cos\omega t = \frac{\sqrt{3}}{2}A,$

$\cos\omega t = \frac{\sqrt{3}}{2}, \qquad \omega t = \frac{\pi}{6} = \frac{2\pi}{T}\times t$

$\Rightarrow t = \frac{T}{12}, \qquad v_{av} = \frac{\sqrt{3}A/2}{T/12} = \frac{\sqrt{3}A}{2}\times\frac{12}{T} = \frac{6\sqrt{3}A}{T}$

04. $V = -\frac{GM}{R+h}, \quad g = \frac{GM}{R^2}, \qquad GM = gR^2,$

$V = -\frac{gR^2}{R+h} = -\frac{gR}{\left(1+\frac{h}{R}\right)} = -gR\left(1+\frac{h}{R}\right)^{-1}$

$= -gR\left(1-\frac{h}{R}\right) = -g(R-h)$

05. $\phi = h\upsilon = 6.63\times10^{-34}\times7.5\times10^{14}$

$= 49.725\times10^{-20}$ J

$\phi = \frac{49.725\times10^{-20}}{1.6\times10^{-19}} = 3.1$ eV.

07. $C = A\cdot B \quad \Rightarrow$ AND gate

09. $\vec{\tau} = \vec{r}\times\vec{F} = 0 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & -4 \\ \alpha & 4 & 8 \end{vmatrix}$

$= i(-16+16) - \hat{j}(24+4\alpha) + k(12+2\alpha)$

$\Rightarrow 24 + 4\alpha = 12 + 2\alpha = 0 \quad \Rightarrow \alpha = -6$

10. $nR\Delta T = 150$ J,

$\Delta Q = nC_P\Delta T = n\times\frac{7R}{2}\Delta T = \frac{7}{2}\times150$

$= 7\times75 = 525$ J,

$\Delta Q = 525$ J.

11. $v = u - gt, \qquad 0 = u - gt,$

$u = gt = 10\times6 = 60$ m/s

12. $q_1 = q_2 = \frac{5.4-1.4}{2} = \frac{4}{2} = 2$ nC,

$F = \frac{kq_1q_2}{r^2} = \frac{9\times10^9\times2\times10^{-9}\times2\times10^{-9}}{(0.2)^2}$

$= \frac{9\times4\times10^{-9}}{4\times10^{-2}} = 9\times10^{-7}$ N

$\Rightarrow x = 9$

13. $\sigma = e(\mu_e n_e + \mu_n n_n)$

$= 1.6\times10^{-19}(3\times8\times10^{18} + 0.02\times8\times10^{19})$

$= 1.6\times10^{-19}(24\times10^{18} + 0.16\times10^{19})$

$= 1.6\times10^{-19}(2.4\times10^{19} + 0.16\times10^{19})$

$= 1.6(2.4 + 0.16 = 2.56) = 4.096 \approx 4.01\ (\Omega\text{-m})^{-1}$

14. Pressure gradient $= \left[\frac{P}{L}\right] = \left[\frac{ML^{-1}T^{-2}}{L}\right] = [ML^{-2}T^{-2}] \Rightarrow (A\to III)$

Energy density $= \left[\frac{E}{V}\right] = \left[\frac{ML^2T^{-2}}{L^3}\right] = [ML^{-1}T^{-2}] \Rightarrow (B\to II)$

Electric field $= \left[\frac{F}{q}\right] = \left[\frac{F}{It}\right] = \left[\frac{MLT^{-2}}{AT}\right] = [MLT^{-3}A^{-1}] \Rightarrow (C\to IV)$

Latent heat $= \left[\frac{Q}{M}\right] = \left[\frac{ML^2T^{-2}}{M}\right] = [M^0L^2T^{-2}] \Rightarrow (D\to I)$

$(A\to III), (B\to II), (C\to IV), (D\to I)$

15. $v_T = \frac{2r^2(d-\rho)g}{9\eta}$

$\Rightarrow v_T \propto r^2,$

$27\times\frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3, \qquad R = 3r$

$\frac{v_1}{v_2} = \frac{r_1^2}{r_2^2}, \qquad \frac{18}{v_2} = \frac{r_1^2}{(3r_1)^2} = \frac{1}{9},$

$v_2 = 18\times9 = 162$ cm/sec.

16. Coercivity $= nI = \frac{N}{l}I = \frac{500}{0.25}\times2$

$= 4000\frac{A}{m}$

18. $v_1 = \frac{m_1 - m_2}{m_1 + m_2} u$, $v_2 = \frac{2m_1}{m_1 + m_2} u$, $v_1 = -v_2$

$\Rightarrow \frac{m_2 - m_1}{m_1 + m_2} = \frac{2m_1}{m_1 + m_2}$

$\Rightarrow m_2 - m_1 = 2m_1$, $m_2 = 3m_1$, $\frac{m_1}{m_2} = \frac{1}{3}$

19. $\upsilon_1 = \frac{n}{2l}\sqrt{\frac{T}{\mu}}$, $\upsilon_2 = \frac{n+1}{2l}\sqrt{\frac{T}{\mu}}$, $\upsilon_2 - \upsilon_1 = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$

$80 = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$, $80 \times 2 \times 0.4 = 64 = \sqrt{\frac{T}{\mu}}$, $64^2 = \frac{3200}{\mu}$

$\mu = \frac{3200}{64 \times 64} = \frac{100}{2 \times 64} = \frac{25}{32}$

$= 0.78125 \text{ kg/m} \approx 0.78 \text{ kg/m}$

20. $Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{300^2 + (600 - 200)^2}$

$= \sqrt{300^2 + 400^2} = 500$

$\Rightarrow Z = 500\ \Omega$, $I_{rms} = \frac{V_{rms}}{Z} = \frac{240}{500} = \frac{12}{25} = 0.48\text{ A}$

21. $\mu \times 16 + \mu \times 8 = 40$, $24\mu = 40$,

$\mu = \frac{40}{24} = \frac{5}{3} = 1.67$

24. $d \sin\theta = 3\lambda$, $\sin\theta = \frac{3\lambda}{d} = \frac{3 \times 4 \times 10^{-7}}{24 \times 10^{-7}} = \frac{1}{2}$,

$\theta = \frac{\pi}{6}$

25. $V \propto \frac{1}{r}$ and $E \propto \frac{1}{r^2}$

[NCERT Class - XII, Part - I]

26. All curves approach the ideal gas behaviour for low pressures and high temperatures.

PV = constant i.e. $P \propto \frac{1}{V}$ (Boyle's law)

$\Rightarrow T_1 > T_2 > T_3$

[NCERT Class - XI, Part - II]

27. $v = 72 \times \frac{5}{18} = 20$ m/s, $\omega = \frac{v}{r} = \frac{20}{0.4} = 50$ rad/sec.

$0 = \omega - \alpha t$, $\omega = \alpha t$, $\alpha = \frac{\omega}{t} = \frac{50}{10} = 5$

$\tau = I\alpha = 4 \times 5 = 20$ N-m.

28. $\frac{dx}{dt} = v_0 + gt + ft^2$, $\int_0^x dx = \int_0^2 (v_0 + gt + ft^2)dt$,

$x = \left[v_0 t + g \times \frac{t^2}{2} + f\frac{t^3}{3} \right]_0^2 = \left[v_0 \times 2 + g \times \frac{4}{2} + \frac{8}{3}f \right]$

$\Rightarrow x = 2v_0 + 2g + \frac{8}{3}f$

29. $Y = \frac{F/A}{\Delta l/l} = \frac{F/A}{\alpha\Delta\theta}$,

$\Delta\theta = \frac{F/A}{\alpha \times Y} = \frac{4 \times 10^8}{2 \times 10^{-5} \times 12 \times 10^{10}} = \frac{1}{6} \times 10^3 = 167°C$,

$200 - \theta = 166.67$, $\theta = 200 - 166.67 = 33.33$

$\Rightarrow \theta = 33.33\ °C$

30. $\rho gh = 4P - P = 3P$, $\rho g \times \frac{2h}{3} = \frac{2}{3} \times 3P = 2P$,

Pressure at the bottom of the tank = 2P + P = 3P

31. $\delta = 2i - A$, $i = e = \frac{2}{3}A$,

$\delta = 2 \times \frac{2A}{3} - A = \frac{4A}{3} - A = \frac{A}{3}$

$\Rightarrow \delta = \frac{60°}{3} = 20°$

32. $E_3 = -\frac{13.6}{3^2} = -\frac{13.6}{9} = -1.51$ eV,

$\Delta E = 18 - 1.51 = 16.49$ eV

33. $y_{6th} = \frac{D}{d} \times 6\lambda$, $y_1 = \frac{D}{d} \times \frac{\lambda}{2}$

$\Delta y = y_{6th} - y_1 = \frac{D}{d}\left(6\lambda - \frac{\lambda}{2}\right) = \frac{D}{d} \times \frac{11}{2}\lambda$

$= 3.96 \times 10^{-3}$,

$\frac{1.2}{10^{-3}} \times 5.5 \times \lambda = 3.96 \times 10^{-3}$

$\Rightarrow \lambda = \frac{3.96 \times 10^{-3} \times 10^{-3}}{5.5 \times 1.2} = \frac{3.96 \times 10^{-6}}{6.6}$

$= 0.6 \times 10^{-6} = 6 \times 10^{-7}$ m

$\Rightarrow \lambda = 6000$ Å

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34. $\eta = \frac{\text{output power}}{\text{input power}} = \frac{e_S I_S}{e_P I_P} = 0.8 = \frac{240 \times I_S}{2400 \times 6}$

$\Rightarrow I_S = \frac{0.8 \times 2400 \times 6}{240} = 48$ A.

35. $\vec{F} = q\,\vec{v} \times \vec{B}$ $\Rightarrow$ North

36. $2\pi r_{4th} = 4\lambda$

37. $I_g G = (I - I_g)s, \quad 5 \times 40 = (85 - 5) \times s = 80\ s,$

$s = \frac{5 \times 40}{80} = 2.5$

$\Rightarrow s = 2.5\ \Omega.$

43. $mg = 8g \sin 30° + \mu \times 8g \cos 30°$

$= 4g + \frac{1}{2\sqrt{3}} \times 8g \times \frac{\sqrt{3}}{2} = 4g + 2g = 6g$

$\Rightarrow mg = 6g, \quad m = 6$ kg.

45. $T = \frac{2u\sin\theta}{g}, \quad H = \frac{u^2 \sin^2\theta}{2g}, \quad R = \frac{u^2 \sin 2\theta}{g}$

$\Rightarrow$ (II, IV)

BIOLOGY

46. NCERT class 11th (page no 20, 21)
47. [NCERT (Class-11)Pg. 203]
48. NCERT class 11th (page no. 95)
49. [NCERT (Class-12)Pg. 31]
50. NCERT class 11th (page no 7)
51. [NCERT (Class-11)Pg. 224,226]
52. NCERT class 11th (page no 124)
53. [NCERT (Class-11)Pg. 101]
54. NCERT class 11th (page no 118, 158)
55. [NCERT (Class-12)Pg. 34]
56. NCERT class 12th (page no 74)
57. [NCERT (Class-11)Pg. 189]
58. [NCERT (Class-11)Pg. 241,212]
59. NCERT class 12th (page no 59, 64)
60. NCERT class 11th (page no 67)
62. [NCERT (Class-12)Pg. 35]
63. NCERT class 11th (page no 124)
64. NCERT class 11th (page no 128)
65. [NCERT (Class-12)Pg. 46]
66. [NCERT (Class-11)Pg. 188]
67. NCERT class 11th (page no 126)
68. [NCERT (Class-11)Pg. 183]
69. NCERT class 12th (page no 99)
70. [NCERT (Class-11)Pg. 209]
72. [NCERT (Class-11)Pg. 245,246]
73. NCERT class 11th (page no 167)
74. [NCERT (Class-12)Pg. 131,134]
75. NCERT class 11th (page no 76)
76. [NCERT (Class-11)Pg. 237]
77. Old NCERT calss 11th (page no 67)
78. [NCERT (Class-12)Pg. 36,37]
79. NCERT class 11th (page no 14)
80. NCERT class 11th (page no 24)
81. [NCERT (Class-11)Pg. 48]
82. NCERT class 11th (page no 159)
83. [NCERT (Class-11)Pg. 184]
84. NCERT class 12th (page no 74, 75)
85. [NCERT (Class-11)Pg. 234]
87. [NCERT (Class-12)Pg. 170]
88. NCERT class 12th (page no 94)
89. [NCERT (Class-12)Pg. 153]
90. NCERT class 11th
91. [NCERT (Class-12)Pg. 171]
92. NCERT class 11th (page no 87)
93. [NCERT (Class-12)Pg. 122,113,111,115]
94. NCERT class 11th (page no 135)
95. [NCERT (Class-11)Pg. 140]
97. [NCERT (Class-12)Pg. 151]
98. Old NCERT class 11th (page no 78)
99. [NCERT (Class-11)Pg. 197,200,203,202]
100. NCERT class 11th (page no 176)
101. NCERT class 12th (page no 104)
102. NCERT class 11th (page no 29)
103. [Old NCERT (Class-11)Pg. 114]
104. NCERT class 11th (page no 96)
105. [NCERT (Class-12)Pg. 138]
106. NCERT class 12th (page no 76)
107. [NCERT (Class-12)Pg. 168]
108. [NCERT (Class-11)Pg. 82]
109. [NCERT (Class-11)Pg. 218]
111. [NCERT (Class-11)Pg. 206]
112. NCERT class 11th (page no 7)
113. [NCERT (Class-12)Pg. 122]
114. NCERT class 12th (page no 95)
115. [NCERT (Class-11)Pg. 45]

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116. NCERT class 12th (page no 224)
117. [NCERT (Class-11)Pg. 241]
118. [NCERT (Class-12)Pg. 178]
119. NCERT class 12th (page no 19)
120. [NCERT (Class-11)Pg. 209]
121. OLD NCERT class 11th (page no 86 – 87)
122. [NCERT (Class-12)Pg. 169]
123. NCERT class 11th (page no 109)
124. [NCERT (Class-12)Pg. 171]
126. [NCERT (Class-12)Pg. 44]
128. NCERT class 12th (page no 212, 213)
129. [NCERT (Class-11)Pg. 222]
130. NCERT class 12th (page no 201)
131. NCERT class 12th (page no 185)
132. NCERT class 12th (page no 208)
133. NCERT class 11th (page no 3)
134. [NCERT (Class-11)Pg. 45]
135. [NCERT (Class-12)Pg. 142]

CHEMISTRY

136. $n_{I_2} = 0.2$, $n_{Benzene} = 0.8$

mass of solution = $0.2 \times 254 + 0.8 \times 78$

$= 113.2$ g

$$\text{Volume of solution} = \frac{113.2}{d}\text{ mL}$$

$$= \frac{0.1132}{d}\text{ L}$$

$$\text{Molarity} = \frac{\text{No of mole of } I_2}{\text{Volume of solution (in L)}}$$

$$= \frac{0.2}{0.1132} \times d = 1.77d$$

137. $\log K = -(2000)\frac{1}{T} + 6$ ....(1)

$$\log K = -\frac{E_a}{2.303R}.\frac{1}{T} + \log A \quad ....(2)$$

Compairing (1) & (2)

$\log A = 6$

$\Rightarrow \boxed{A = 10^6}$

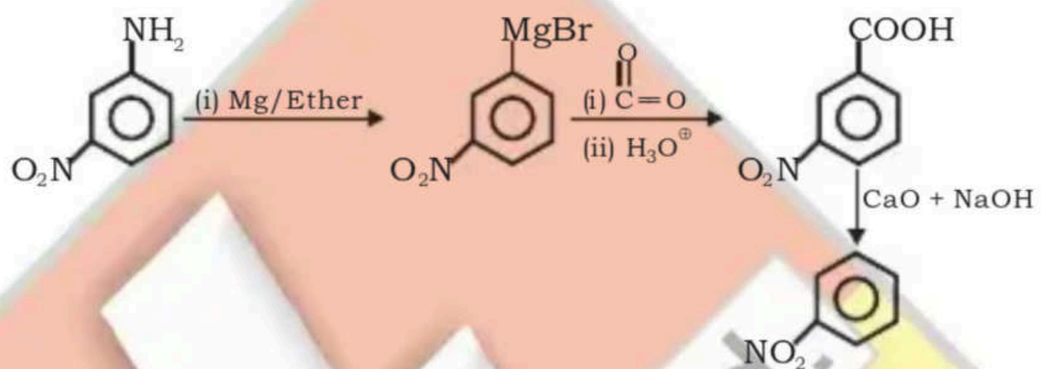
$$\frac{-Ea}{2.303R} = -2000$$

$\Rightarrow E_a = 2.303\ R \times 2000$

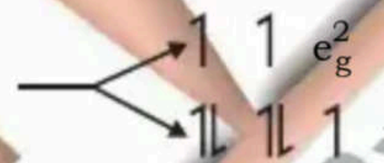
$$= 2.303 \times 8.314 \times 2000 \times \frac{1}{1000}\text{ kJ}$$

$= 38.3$ kJ

138. [NCERT (Pg. 105)]
139. [NCERT (Pg. 287)]
140. [NCERT (Pg. 248)]



141. $Co^{2+} \Rightarrow [Ar]3d^7\ 4s^0$



142. [NCERT (Pg. 109)]
143. $2KMnO_4 + 5H_2C_2O_4 + 6\,HCl \longrightarrow 2MnCl_2 + 10CO_2 + 2KCl + 8H_2O$

Number of electrons involve = 10

$\therefore$ n factor of HCl = $\frac{10}{6} = \frac{5}{3}$

Equivalent mass of HCl = $\frac{3M}{5}$

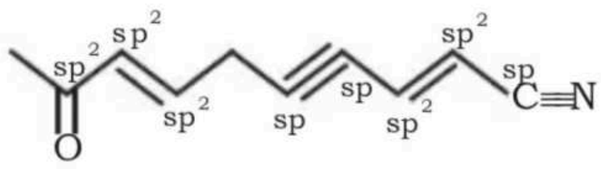
144. $M(s) \longrightarrow M^{2+}(aq) + 2e^-$

$$E_{M/M^{2+}} = E^0_{M/M^{2+}} - \frac{0.059}{2}\log[M^{2+}]$$

$$= 1.18 - \frac{0.059}{2}\log 10^{-1}$$

$$= \left(1.18 + \frac{0.059}{2}\right)V$$

$= 1.2095\ V \approx 1.21\ V$

145. [NCERT (Pg. 236/237)]
146. [NCERT (Pg. 232)]
147. 

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150. $T_b = T_b^0 + i.K_b.m$

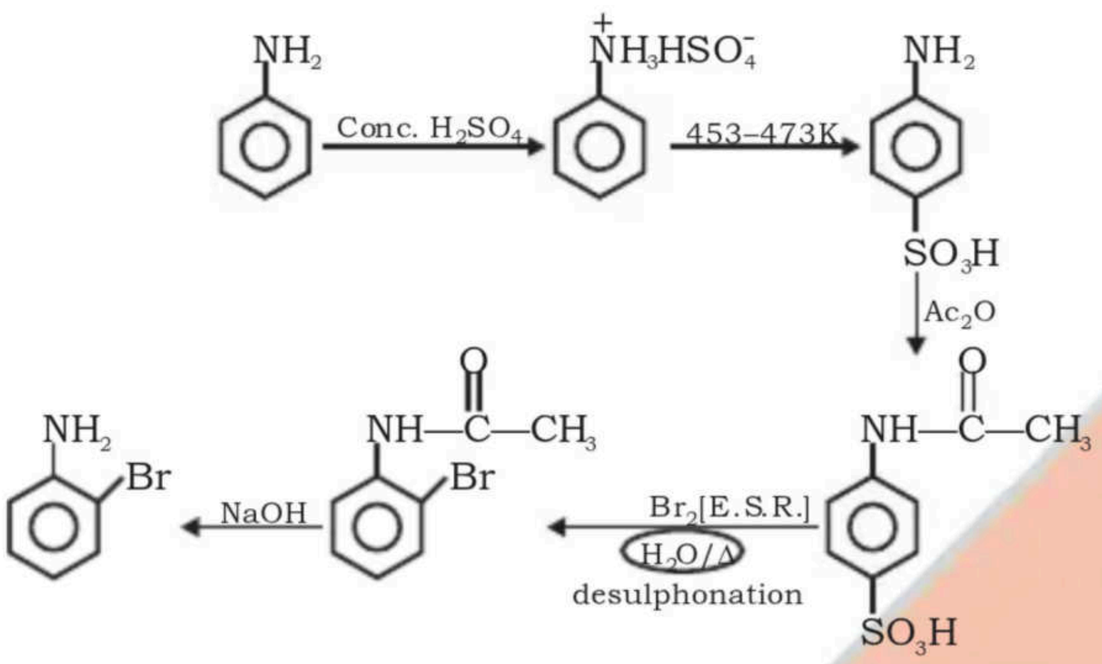
When T_b^0 and K_b are constant

$\boxed{T_b \propto i.m}$

151. [NCERT Option 1 (Class-XII)Pg. 186, Option 2 (Class-XII)Pg. 201, Option 3 (Class-XII)Pg. 198, Option 4 (Class-XI)Pg. 96]

152. [NCERt (Pg. 319)]

154. [NCERT (Pg. 273)]



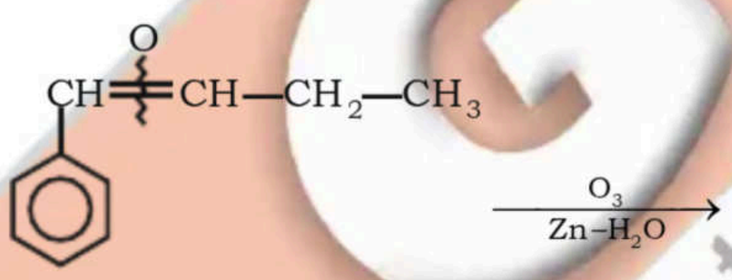
155. Ketone do not give Tollen reagent test but it react with 2,4-DNP.

157. $Fe_2(SO_4)_3 \longrightarrow 2Fe^{3+} + 3SO_4^{2-}$

n-factor = 6

$$\Lambda_{eq} = \frac{\Lambda_m}{n-factor} = \frac{\Lambda_m}{6}$$

158. [NCERT (Pg. 227)]

159. 

160. This haloform reaction.

161. NCERT Page No. - 234

162. $H_2 + I_2 \rightleftharpoons 2HI$

Concentration of H_2 and I_2 decreases until equilbrium condition and concentration of HI increases till equilibrium condition and after equilibrium concentration of all the reactant and products remain constant.

163. Internal energy (U) is a state function.

Change in state function for a cyclic process is zero.

$\therefore$ ΔU (Case-I) = ΔU (Case-II) = ΔU (Case-III) = 0

164. $$\frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{n_1^2} - \frac{1}{n_2^2} = \frac{1}{\lambda.R_H.Z^2}$$

$$= \frac{1}{900\times10^{-7}\,cm \times 10^5\,cm^{-1} \times 1^2}$$

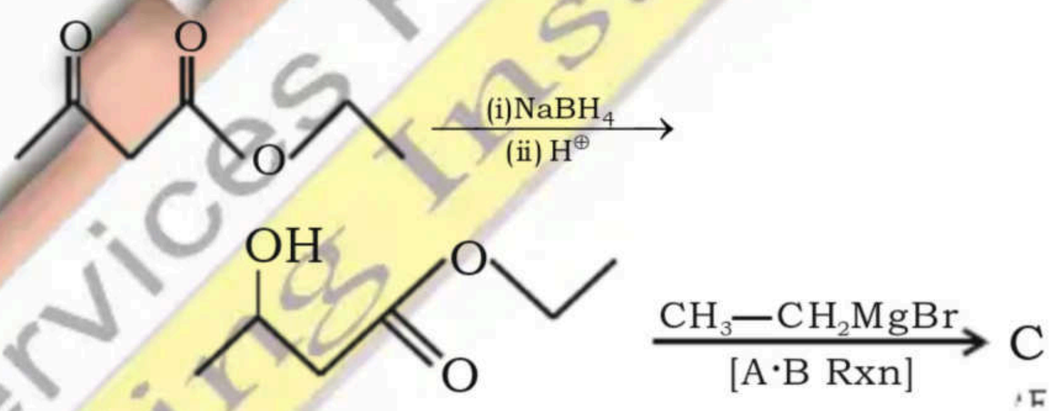
Or, $\boxed{\frac{1}{n_1^2} - \frac{1}{n_2^2} = \frac{1}{9}}$

above relation is correct when,

$n_1 = 3$ and $n_2 = \infty$

i.e. $\boxed{\text{Paschen series}, \infty \rightarrow 3}$

165. NCERT Page No. - 254

166. 

[NCERT Page No. - 257]

167. [NCERT Page No. 389]

168. MA_3B_3 type show fac and mar isomerism.

169. [NCERT Page No. 386/387]

170. [NCERT Page No. 362/363]

171. $$^nCO_2 \text{ (removed)} = \frac{10^{21}}{6.02\times10^{23}} \text{ mol}$$

$$= \frac{10\times10^{20}}{6.02\times10^{23}} \text{ mol}$$

$$= 1.66 \times 10^{-3} \text{ mol}$$

$^nCO_2$ (left) = 2.9×10^{-3} mol

$\therefore$ $^nCO_2$ (taken initially) = $1.66 \times 10^{-3} + 2.9 \times 10^{-3}$

$= 4.56 \times 10^{-3}$ mol

Mass of CO_2 taken initially = $4.56 \times 10^{-3} \times 44$ g

$= 200.64 \times 10^{-3}$ g

$= 200.64$ mg

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173. $\Delta G = \Delta H - T.\Delta S$

at equilibrium, $\Delta G = O$

$$T \rightarrow T_e$$

$$T_e = \frac{\Delta H}{\Delta S}$$

For spontaneous reaction,

$$\Delta G = - Ve$$

and for non-spontaneous reaction,

$$\Delta G = +ve$$

174. [NCERT Page no. - 170]

175. [NCERT Page No. 379)

176. H_2O (l, 27°C, 1 atm) $\longrightarrow$ H_2O (l, 0°C, 1 atm)

$q = m.s.\ \Delta T$ and $\Delta T = 0°C - 27°C = -27$

or, $\Delta H_1 = 18 \times 4.2 \times -27$ J

$= - 2041.2$ J $= - 2.04$ kJ mol^{-1}

H_2O (l, 0°C, 1 atm) $\longrightarrow$ H_2O (S, 0°C, 1 atm)

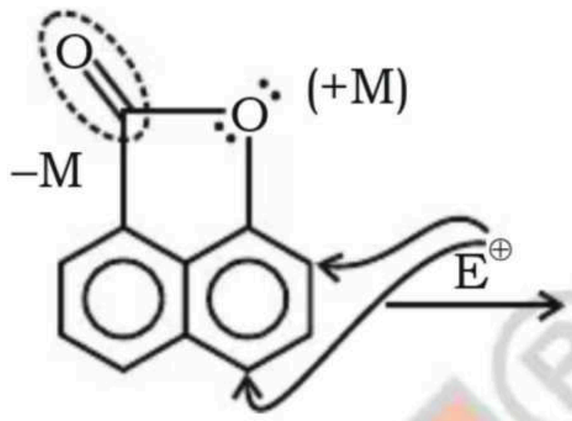
$\Delta H_2 = - 5.96$ kJ mol^{-1}

$\therefore$ H_2O (l, 27°C, 1 atm) $\longrightarrow$ H_2O (S, 0°C, 1 atm)

$\Delta H = \Delta H_1 + \Delta H_2$

$= - 2.04 + (- 5.96) = - 8.00$ kJ mol^{-1}.

177.



The incoming electrophile (Br^+) will attack either ortho or para position with respect to activator group (+M).

178. [NCERT Option A | C | D → Pg. 94 (Class-XI), Option B → Pg. 324 (Class-XII)]

180. [NCERT (Pg. 237)]

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Dear students,

Write your feedback regarding the followings and send it through **message** or **Whatsapp** on **7857072708**.

1. Type of questions quality
2. Conduction of Exam
3. Solution Class
4. Any more facility you need

MAJOR TEST - 04 - 09.03.2025 - ANS. KEY

01.	(3)	**24.**	(2)	**47.**	(2)	**70.**	(2)	**93.**	(4)	**115.**	(1)	**137.**	(4)	**159.**	(1)
02.	(1)	**25.**	(1)	**48.**	(1)	**71.**	(4)	**94.**	(2)	**116.**	(2)	**138.**	(1)	**160.**	(3)
03.	(3)	**26.**	(3)	**49.**	(4)	**72.**	(2)	**95.**	(3)	**117.**	(3)	**139.**	(2)	**161.**	(3)
04.	(4)	**27.**	(4)	**50.**	(3)	**73.**	(3)	**96.**	(4)	**118.**	(2)	**140.**	(1)	**162.**	(2)
05.	(2)	**28.**	(2)	**51.**	(2)	**74.**	(4)	**97.**	(1)	**119.**	(4)	**141.**	(3)	**163.**	(4)
06.	(1)	**29.**	(1)	**52.**	(4)	**75.**	(4)	**98.**	(3)	**120.**	(2)	**142.**	(3)	**164.**	(1)
07.	(2)	**30.**	(3)	**53.**	(3)	**76.**	(4)	**99.**	(1)	**121.**	(3)	**143.**	(1)	**165.**	(2)
08.	(4)	**31.**	(4)	**54.**	(1)	**77.**	(3)	**100.**	(4)	**122.**	(4)	**144.**	(4)	**166.**	(4)
09.	(3)	**32.**	(2)	**55.**	(1)	**78.**	(3)	**101.**	(2)	**123.**	(4)	**145.**	(1)	**167.**	(1)
10.	(4)	**33.**	(1)	**56.**	(3)	**79.**	(4)	**102.**	(4)	**124.**	(3)	**146.**	(3)	**168.**	(1)
11.	(1)	**34.**	(4)	**57.**	(1)	**80.**	(4)	**103.**	(1)	**125.**	(1)	**147.**	(2)	**169.**	(4)
12.	(3)	**35.**	(4)	**58.**	(1)	**81.**	(3)	**104.**	(3)	**126.**	(1)	**148.**	(2)	**170.**	(3)
13.	(2)	**36.**	(2)	**59.**	(4)	**82.**	(1)	**105.**	(3)	**127.**	(4)	**149.**	(4)	**171.**	(1)
14.	(3)	**37.**	(1)	**60.**	(3)	**83.**	(4)	**106.**	(4)	**128.**	(1)	**150.**	(2)	**172.**	(3)
15.	(4)	**38.**	(3)	**61.**	(1)	**84.**	(4)	**107.**	(3)	**129.**	(3)	**151.**	(1)	**173.**	(2)
16.	(1)	**39.**	(2)	**62.**	(3)	**85.**	(4)	**108.**	(3)	**130.**	(2)	**152.**	(2)	**174.**	(4)
17.	(2)	**40.**	(4)	**63.**	(4)	**86.**	(4)	**109.**	(4)	**131.**	(2)	**153.**	(3)	**175.**	(1)
18.	(4)	**41.**	(1)	**64.**	(2)	**87.**	(4)	**110.**	(2)	**132.**	(2)	**154.**	(2)	**176.**	(1)
19.	(2)	**42.**	(4)	**65.**	(4)	**88.**	(1)	**111.**	(4)	**133.**	(3)	**155.**	(3)	**177.**	(3)
20.	(1)	**43.**	(1)	**66.**	(2)	**89.**	(2)	**112.**	(4)	**134.**	(2)	**156.**	(2)	**178.**	(3)
21.	(3)	**44.**	(3)	**67.**	(4)	**90.**	(1)	**113.**	(1)	**135.**	(3)	**157.**	(4)	**179.**	(4)
22.	(4)	**45.**	(2)	**68.**	(2)	**91.**	(2)	**114.**	(1)	**136.**	(1)	**158.**	(1)	**180.**	(1)
23.	(1)	**46.**	(1)	**69.**	(2)	**92.**	(4)								

Major Test - 03 - 02.03.25
Errata
No Any Errta

137. For a first order reaction : A → P, the temperature (T) dependent rate constant (K) was found to follow the equation $\log K = -(2000)\frac{1}{T} + 6.0$. The pre-exponential factor (A) and the activation energy (E_a) respectively are :

(1) $1.0 \times 10^6 s^{-1}$ and 9.2 kJ mol^{-1}

(2) 6.0 s^{-1} and 16.6 kJ mol^{-1}

(3) $1.0 \times 10^6 s^{-1}$ and 16.6 kJ mol^{-1}

(4) $1.0 \times 10^6 s^{-1}$ and 38.3 kJ mol^{-1}

137. $\log K = -(2000)\frac{1}{T} + 6$ ....(1)

$$\log K = = -\frac{E_a}{2.303R}.\frac{1}{T} + \log A \quad ....(2)$$

Compairing (1) & (2)

$\log A = 6$

$\Rightarrow \boxed{A = 10^6}$

$$\frac{-Ea}{2.303R} = -2000$$

$\Rightarrow E_a = 2.303\ R \times 2000$

$$= 2.303 \times 8.314 \times 2000 \times \frac{1}{1000} kJ$$

$= 38.3$ kJ

139. Which one of the follolwing compound contains α-1-4 glycosidic linkage ?

(1) Sucrose **(2) Maltose**

(3) Lactose (4) Cellulose

139. [NCERT (Pg. 287)]

145. The number of element from the following that do not belong to actinoids is :

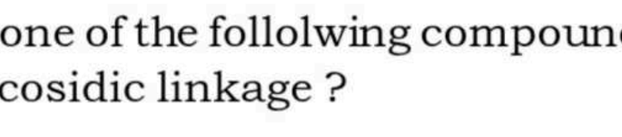
Er, Cm, Ho, Md, No, Tm and Dy

(1) 4 (2) 6

(3) 5 (4) 3

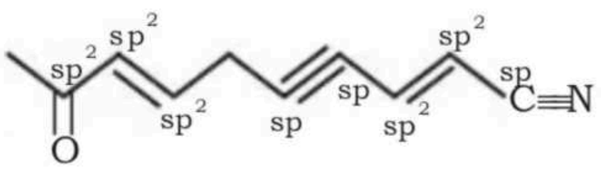
145. [NCERT (Pg. 236/237)]

147. In the given structure number of sp and sp^2 hybridized carbon atoms present respectively are :



(1) 3 and 6 **(2) 3 and 5**

(3) 4 and 6 (4) 4 and 5

147. 

151. In which of the following arrangements, the sequence is not according to the property written against it :

(1) O > S > Se > Te : Boiling point

(2) HI > HBr > HCl > HF : Acidic strength

(3) $Cl_2 > Br_2 > F_2 > I_2$: Bond dissociation enthalpy

(4) N > O > C > B : First ionisation enthalpy

151. [NCERT Option 1 (Class-XII)Pg. 186, Option 2 (Class-XII)Pg. 201, Option 3 (Class-XII)Pg. 198, Option 4 (Class-XI)Pg. 96]

158. Which of the ions from the following liberate hydrogen from a dilute acid is ?

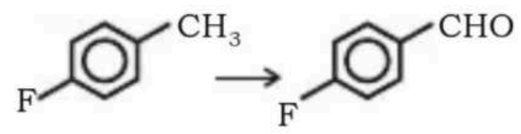
(1) V^{2+} (2) Co^{3+}

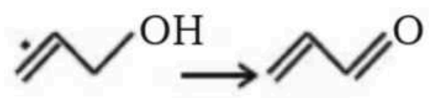
(3) Cr^{2+} (4) Fe^{+2}

158. [NCERT (Pg. 227)]

161. Match the Column-I with Column-II

A. $CH_3CN \longrightarrow CH_3CHO$ i. P.C.C.

B.  ii. $CrO_2Cl_2;H_2O$

C.  iii. O_3/H_2O–Zn

D.  iv. DiBAL–H, $H_2O^{\oplus}$

(1) A → i, B → ii, C → iii, D → iv

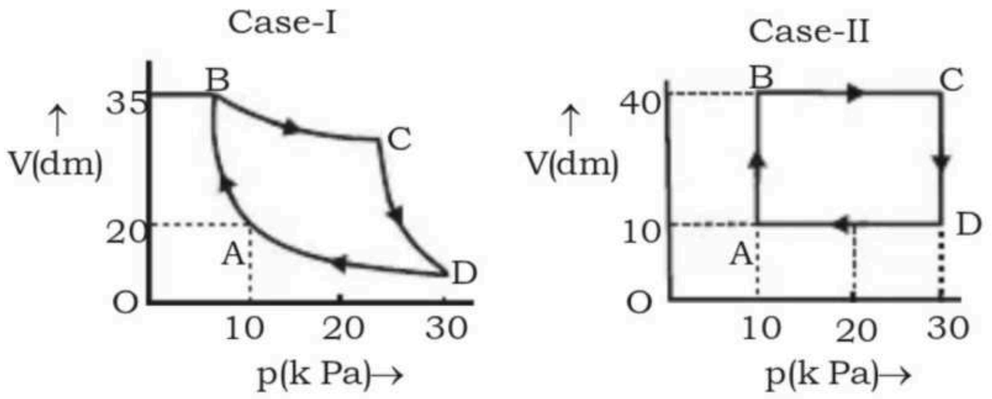
(2) A → iii, B → ii, C → i, D → iv

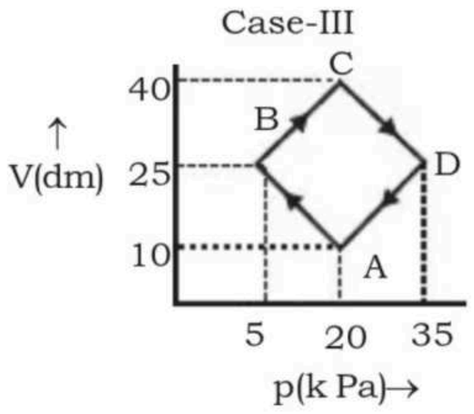
(3) A → iv, B → ii, C → i, D → iii

(4) A → i, B → iv, C → iii, D → ii

161. NCERT Page No. - 234

163. An ideal gas undergoes a cyclic transformation





starting from the point A and coming back to the same point by tracing the path A→B→C→D→A as shown in the three cases above.

Choose the correct option regarding ΔU .

(1) ΔU (Case-III) > ΔU (Case-II) > ΔU (Case-I)

(2) ΔU (Case-I) > ΔU (Case-II) > ΔU (Case-III)

(3) ΔU (Case-I) > ΔU (Case-III) > ΔU (Case-II)

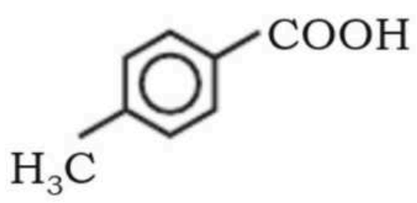
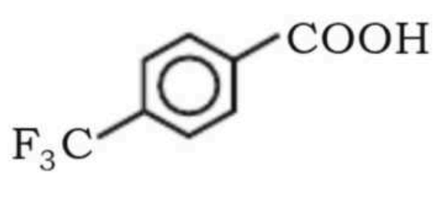
(4) ΔU (Case-I) = ΔU (Case-II) = ΔU (Case-III)

163. Internal energy (U) is a state function.

Change in state function for a cyclic process is zero.

∴ ΔU (Case-I) = ΔU (Case-II) = ΔU (Case-III) = 0

165. In which of the following I is more acidic than II.

	I	II
(1)	CH_3CO_2H	CH_2FCO_2H
(2)	**CH_2FCO_2H**	**CH_2ClCO_2H**
(3)	$CH_3CHFCH_2CO_2H$	$CH_2FCH_2CO_2H$
(4)		

165. NCERT Page No. - 254

170. The incorrect matching pair is

Column-I [Elements]	Column-II [PPt color in Lassiagne test]
(1) Nitrogen	→ Prussian Blue
(2) Sulphur	→ Violet
(3) Chlorine	**→ Blue**
(4) Phosphorus	→ Yellow

170. [NCERT Page No. 362/363]

173. The effect of temperature on spontaneity of reactions are represented as :

	ΔH	ΔS	Temperature	Spontaneity
(A)	+	–	any T	Non spontaneous
(B)	+	+	$T < T_e$	Spontaneous
(C)	–	–	$T < T_e$	Nonspontaneous
(D)	–	+	any T	spontaneous

Select correct ones :

(1) (B) and (D) only **(2) (A) and (D) only**

(3) (B) and (C) only (4) (A) and (C) only

173. $\Delta G = \Delta H - T.\Delta S$

at equilibrium, $\Delta G = O$

$T \rightarrow T_e$

$$T_e = \frac{\Delta H}{\Delta S}$$

For spontaneous reaction,

$$\Delta G = -Ve$$

and for non-spontaneous reaction,

$$\Delta G = +ve$$

176. Calculate the enthalpy change when 1 mole of water at 27°C and 1 atm pressure is converted to ice at 0°C. Given the enthalpy of fusion of ice is 5.96 kJ mol^{-1} and heat capacity of water is 4.2 $J°c^{-1}g^{-1}$.

(1) – 8.00 kJ mol^{-1} (2) 6.36 kJ mol^{-1}

(3) – 5.96 kJ mol^{-1} (4) 5.96 kJ mol^{-1}

176. H_2O (l, 27°C, 1 atm) ⟶ H_2O (l, 0°C, 1 atm)

q = m.s. ΔT and ΔT = 0°C – 27°C = –27

or, $\Delta H_1 = 18 \times 4.2 \times -27$ J

$= -2041.2$ J $= -2.04$ kJ mol^{-1}

H_2O (l, 0°C, 1 atm) ⟶ H_2O (S, 0°C, 1 atm)

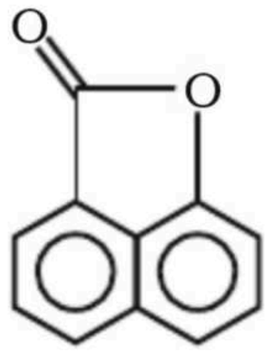
$\Delta H_2 = -5.96$ kJ mol^{-1}

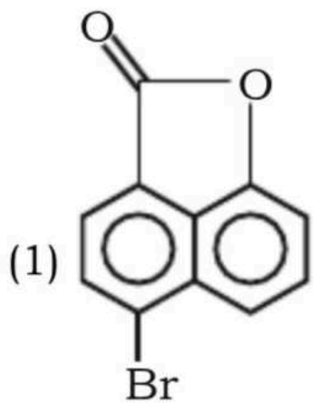
∴ H_2O (l, 27°C, 1 atm) ⟶ H_2O (S, 0°C, 1 atm)

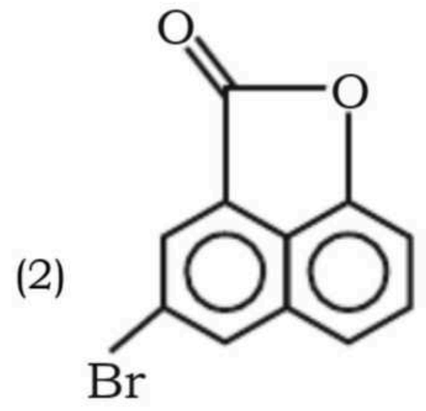
$\Delta H = \Delta H_1 + \Delta H_2$

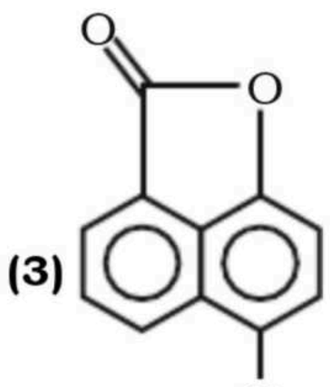
$= -2.04 + (-5.96) = -8.00$ kJ mol^{-1}.

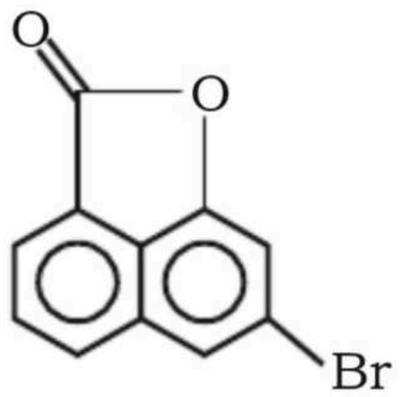
177. If the given compound is subject to halogenation (Br_2/Fe) the major product formed is

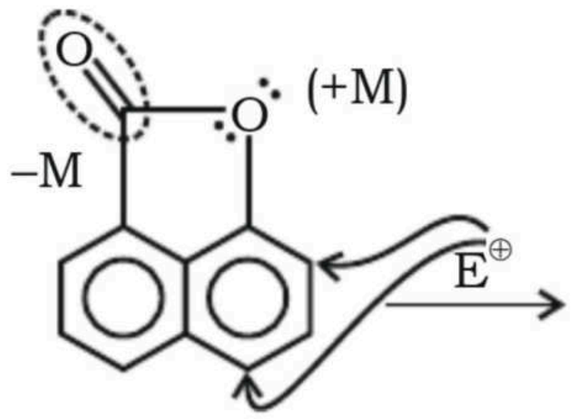


(1) 

(2) 

(3) 

(4) 

177. 

The incoming electrophile (Br^+) will attack either ortho or para position with respect to activator

group (+M).

179. Which of the following are not an isostructural pair ?

(1) XeF_2, IF_2^- (2) SiF_4, PCl_4^+

(3) SO_4^{2-}, CrO_4^{2-} **(4)** NH_3, NO_3^-

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